

ESS og MAX-IV, neutron-instrumentering

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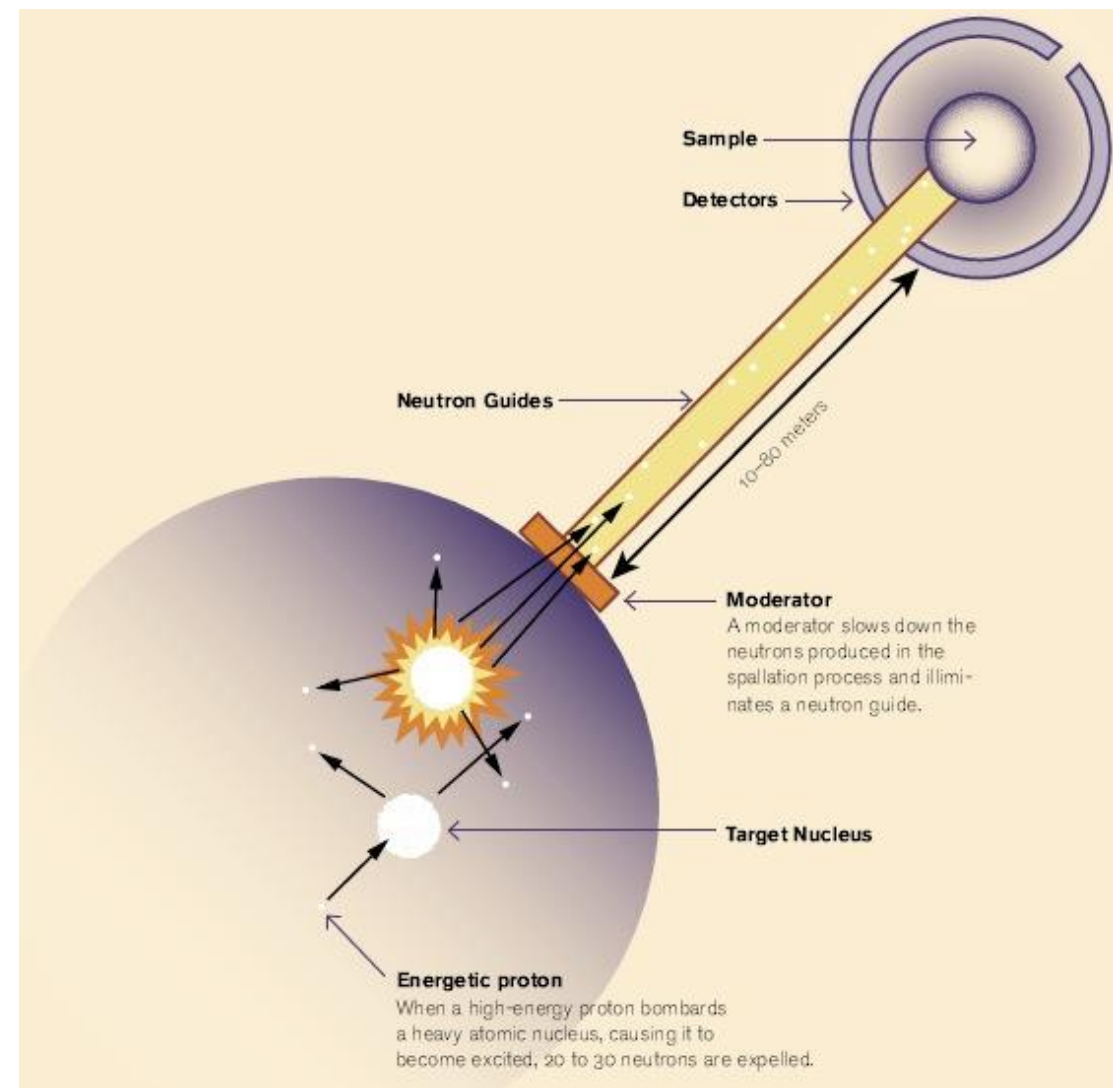
Ken Andersen, ESS

Roger Pynn, Indiana University

Sunil K. Sinha, UCSD

Thom Mason, ORNL

Colin Carlile, Uppsala Universitet (tidl. ESS)



Neutroner - der kan bruges til noget...

1. Det store overblik

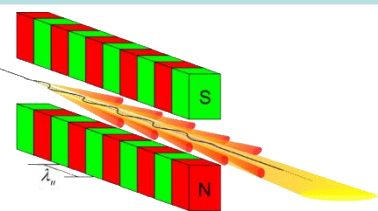
- ESS og MAX-V
- Røntgen vs. Neutroner
- Lidt mere i dybden med ESS

2. Moderne neutron-instrumentering

- At køle neutroner - til relevant bølgelængde / energi
- Transport over afstand
- Tilpassede beams til det enkelte eksperiment

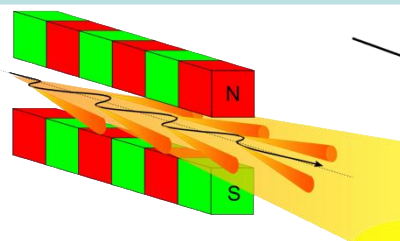
3. ESS som projekt

ESS og MAX-IV I Lund, stråling via partikel-acceleration:

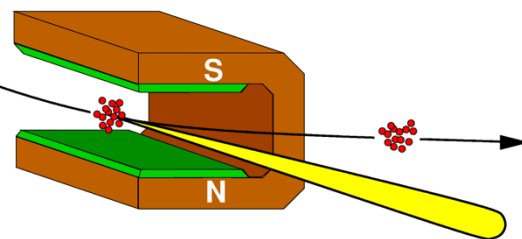


Undulator

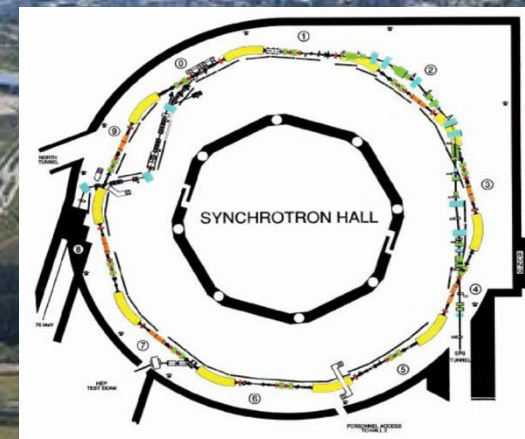
(Insertion Devices)



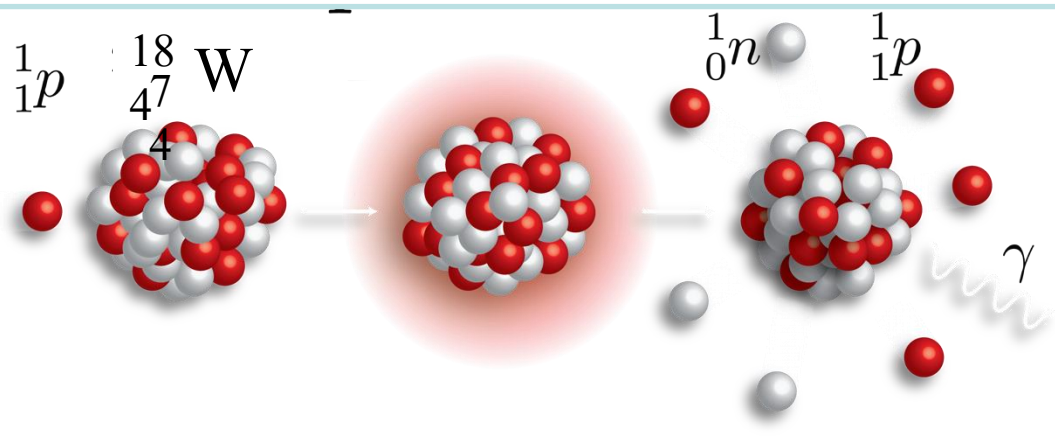
Wiggler



Bending Magnet



MAX-IV



neutroner

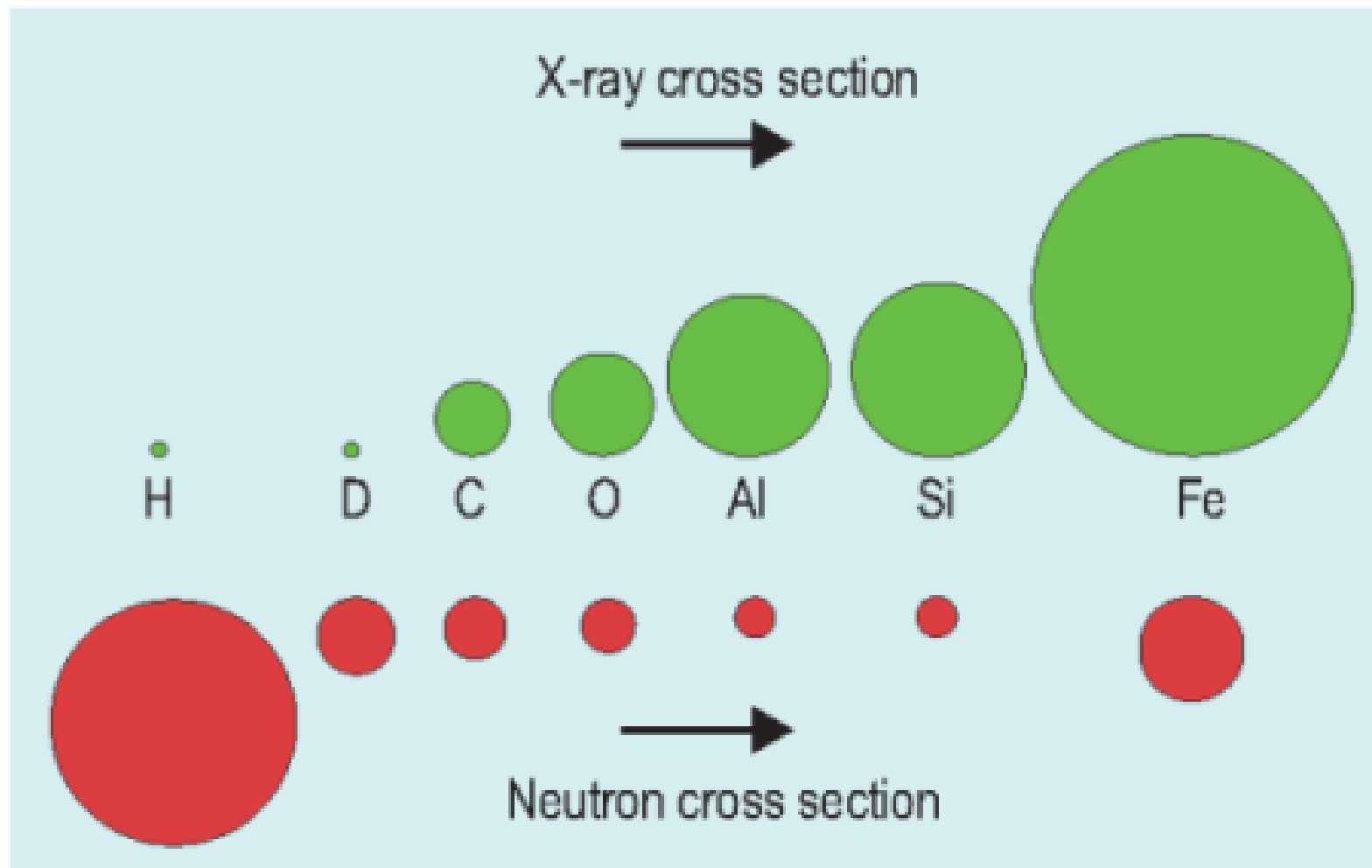
Target



EUROPEAN
SPALLATION
SOURCE

Både neutroner og X-rays vekselvirker med atomer

- X-rays "ser" elektronskyen rundt om atomkernen - høyere atomnummer, mere spredning
- Neutroner "ser" derimod kernen - ingen simpel sammenheng med atomnummer



Periodic Table of the Elements

Ground State Electron Configurations

1A

2A

1

H

1s¹

2

He

1s²

3

Li

1s²2s¹

4

Be

1s²2s²

11

Na

[Ne]3s¹

12

Mg

[Ne]3s²

13

Al

[Ne]3s²3p¹

14

Si

[Ne]3s²3p²

15

P

[Ne]3s²3p³

16

S

[Ne]3s²3p⁴

17

Cl

[Ne]3s²3p⁵

18

Ar

[Ne]3s²3p⁶

19

K

[Ar]4s¹

20

Ca

[Ar]4s²

21

Sc

[Ar]3d¹4s²

22

Ti

[Ar]3d²4s²

23

V

[Ar]3d³4s²

24

Cr

[Ar]3d⁵4s¹

25

Mn

[Ar]3d⁵4s²

26

Fe

[Ar]3d⁶4s²

27

Co

[Ar]3d⁷4s²

28

Ni

[Ar]3d⁸4s²

29

Cu

[Ar]3d¹⁰4s¹

30

Zn

[Ar]3d¹⁰4s²

31

Ga

[Ar]3d¹⁰4s²4p¹

32

Ge

[Ar]3d¹⁰4s²4p²

33

As

[Ar]3d¹⁰4s²4p³

34

Se

[Ar]3d¹⁰4s²4p⁴

35

Br

[Ar]3d¹⁰4s²4p⁵

36

Kr

[Ar]3d¹⁰4s²4p⁶

37

Rb

[Kr]5s¹

38

Sr

[Kr]5s²

39

Y

[Kr]4d¹5s²

40

Zr

[Kr]4d²5s²

41

Nb

[Kr]4d⁴5s¹

42

Mo

[Kr]4d⁵5s¹

43

Tc

[Kr]4d⁵5s²

44

Ru

[Kr]4d⁷5s¹

45

Rh

[Kr]4d⁸5s¹

46

Pd

[Kr]4d¹⁰

47

Ag

[Kr]4d¹⁰5s¹

48

Cd

[Kr]4d¹⁰5s²

49

In

[Kr]4d¹⁰5s²5p¹

50

Sn

[Kr]4d¹⁰5s²5p²

51

Sb

[Kr]4d¹⁰5s²5p³

52

Te

[Kr]4d¹⁰5s²5p⁴

53

I

[Kr]4d¹⁰5s²5p⁵

54

Xe

[Kr]4d¹⁰5s²5p⁶

55

Cs

[Xe]6s¹

56

Ba

[Xe]6s²

57-71

Lanthanides

72

Hf

[Xe]4f¹⁴5d²6s²

73

Ta

[Xe]4f¹⁴5d³6s²

74

W

[Xe]4f¹⁴5d⁴6s²

75

Re

[Xe]4f¹⁴5d⁵6s²

76

Os

[Xe]4f¹⁴5d⁶6s²

77

Ir

[Xe]4f¹⁴5d⁷6s²

78

Pt

[Xe]4f¹⁴5d⁹6s¹

79

Au

[Xe]4f¹⁴5d¹⁰6s¹

80

Hg

[Xe]4f¹⁴5d¹⁰6s²

81

Tl

[Xe]4f¹⁴5d¹⁰6s²6p¹

82

Pb

[Xe]4f¹⁴5d¹⁰6s²6p²

83

Bi

[Xe]4f¹⁴5d¹⁰6s²6p³

84

Po

[Xe]4f¹⁴5d¹⁰6s²6p⁴

85

At

[Xe]4f¹⁴5d¹⁰6s²6p⁵

86

Rn

[Xe]4f¹⁴5d¹⁰6s²6p⁶

87

Fr

[Rn]7s¹

88

Ra

[Rn]7s²

89-103

Actinides

104

Rf

[Rn]5f¹⁴6d²7s²

105

Db

[Rn]5f¹⁴6d³7s²

106

Sg

[Rn]5f¹⁴6d⁴7s²

107

Bh

[Rn]5f¹⁴6d⁵7s²

108

Hs

[Rn]5f¹⁴6d⁶7s²

109

Mt

[Rn]5f¹⁴6d⁷7s²

110

Ds

[Rn]5f¹⁴6d⁸7s²

111

Rg

[Rn]5f¹⁴6d⁹7s²

112

Cn

[Rn]5f¹⁴6d¹⁰7s²

113

Uut

[Rn]5f¹⁴6d¹⁰7s²7p¹

114

Fl

[Rn]5f¹⁴6d¹⁰7s²7p²

115

Uup

[Rn]5f¹⁴6d¹⁰7s²7p³

116

Lv

[Rn]5f¹⁴6d¹⁰7s²7p⁴

117

Uus

[Rn]5f¹⁴6d¹⁰7s²7p⁵

118

Uuo

[Rn]5f¹⁴6d¹⁰7s²7p⁶

57

La

[Xe]5d¹6s²

58

Ce

[Xe]4f¹6s²

59

Pr

[Xe]4f²6s²

60

Nd

[Xe]4f³6s²

61

Pm

[Xe]4f⁴6s²

62

Sm

[Xe]4f⁶6s²

63

Eu

[Xe]4f⁷6s²

64

Gd

[Xe]4f⁷5d¹6s²

65

Tb

[Xe]4f⁹6s²

66

Dy

[Xe]4f¹⁰6s²

67

Ho

[Xe]4f¹¹6s²

68

Er

[Xe]4f¹²6s²

69

Tm

[Xe]4f¹³6s²

70

Yb

[Xe]4f¹⁴6s²

71

Lu

[Xe]4f¹⁴5d¹6s²

89

Ac

[Rn]6d¹7s²

90

Th

[Rn]6d²7s²

91

Pa

[Rn]5f²7s²

92

U

[Rn]5f³7s²

93

Np

[Rn]5f⁴7s²

94

Pu

[Rn]5f⁶7s²

95

Am

[Rn]5f⁷7s²

96

Cm

[Rn]5f⁷6d¹7s²

97

Bk

[Rn]5f⁹7s²

98

Cf

[Rn]5f¹⁰7s²

99

Es

[Rn]5f¹¹7s²

100

Fm

[Rn]5f¹²7s²

101

Md

[Rn]5f¹³7s²

102

No

[Rn]5f¹⁴7s²

103

Lr

[Rn]5f¹⁴6d¹7s²

Lanthanides

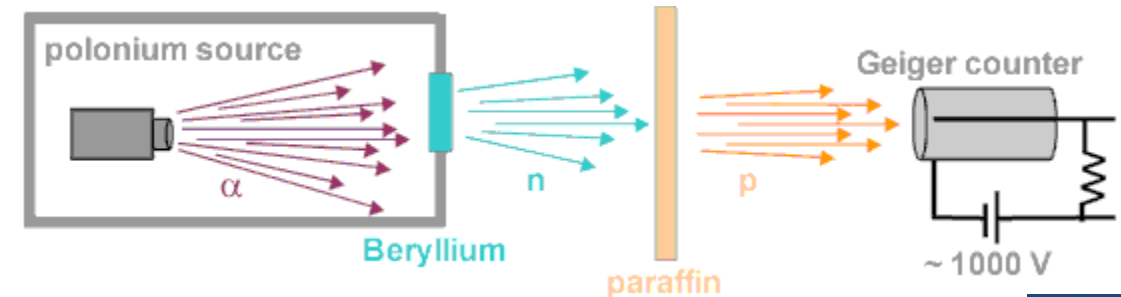
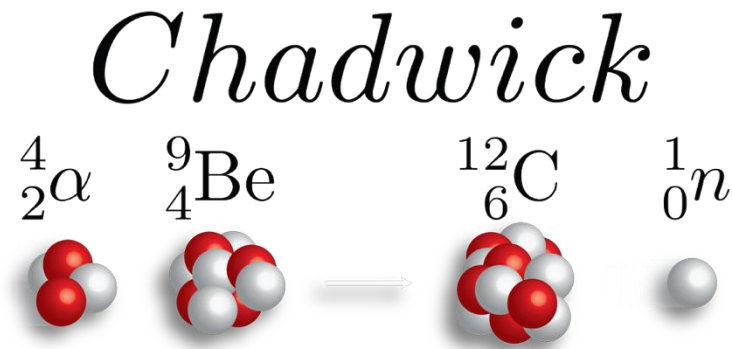
Actinides

* values are based on theory and are not verified

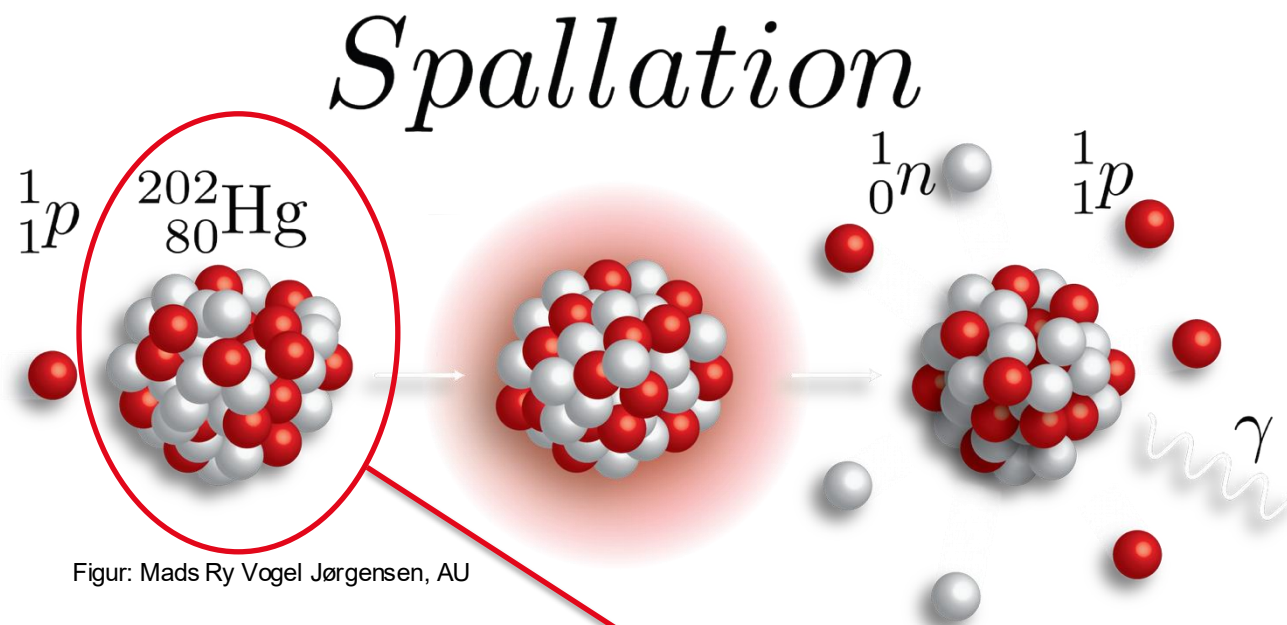
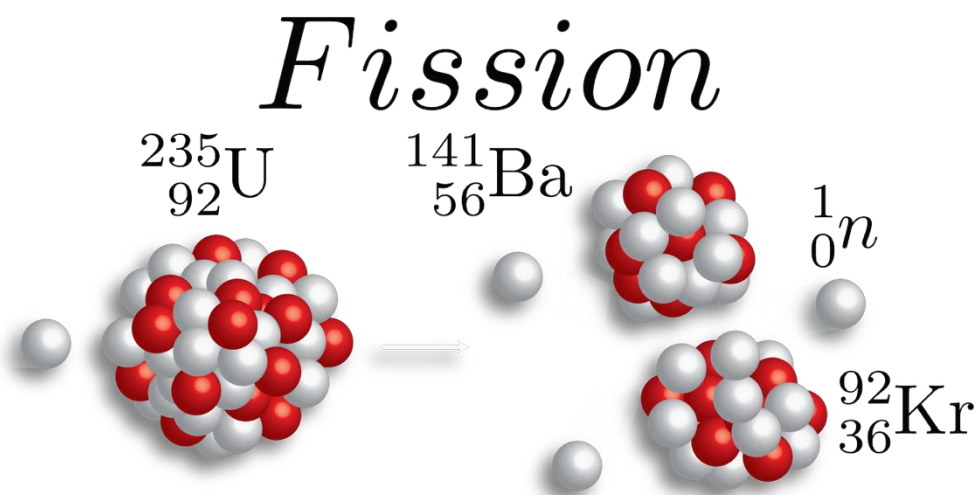
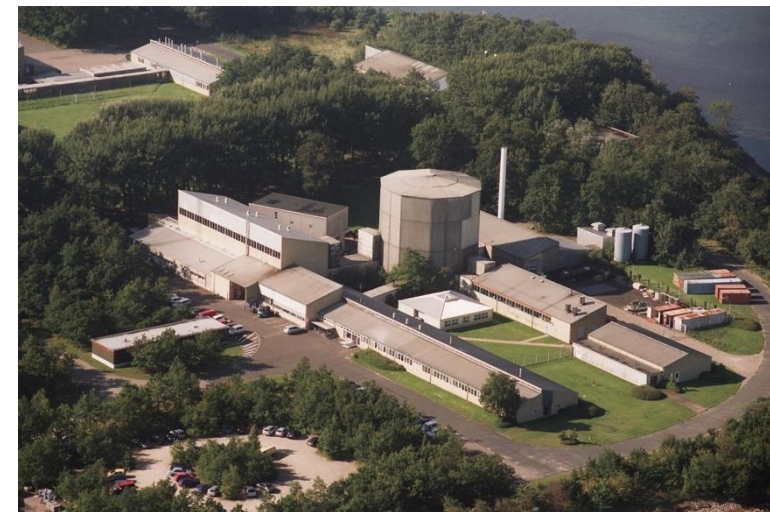
Fig. 2. Neutron and x-ray scattering cross-sections compared. Note that neutrons penetrate through Al much better than x rays do, yet are strongly scattered by hydrogen.

Herfra handler det mest om neutroner :-)

At producere neutroner...

Chadwick's eksperiment / α på Be

RISØ's DR 3 reaktor



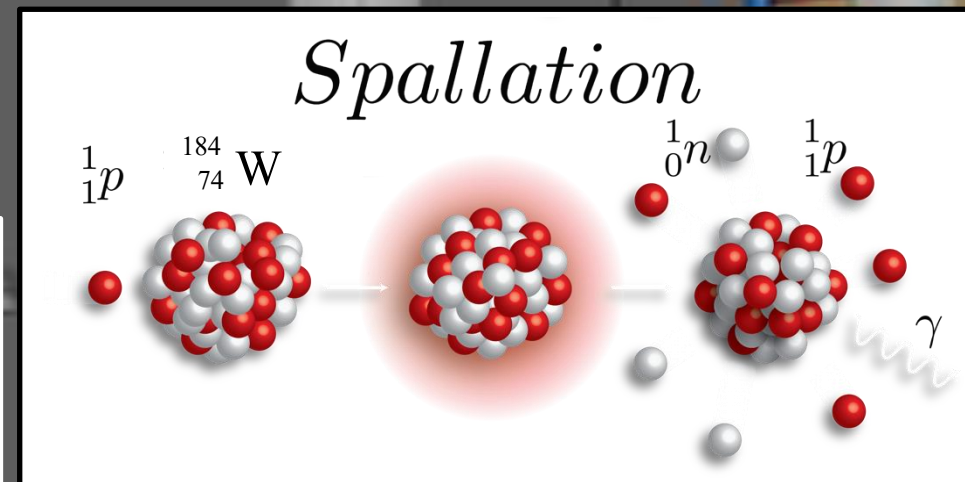
SNS, Oak Ridge Tennessee



Figur: Mads Ry Vogel Jørgensen, AU

ESS target

2.5m tungsten wheel

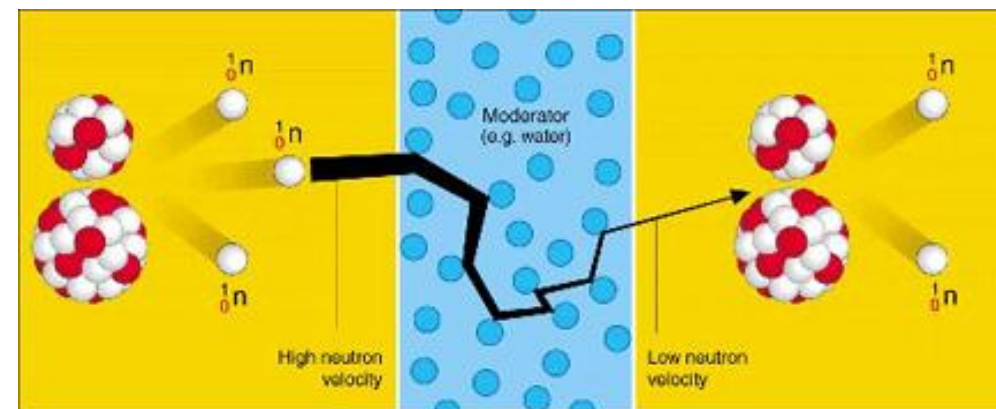


Proton-beam,
3ms pulser @ 14 Hz.
Partikel-energi på sigt 2 GeV.
Beam-effekt på sigt i MW
klassen.

Neutroner

Neutron-moderation

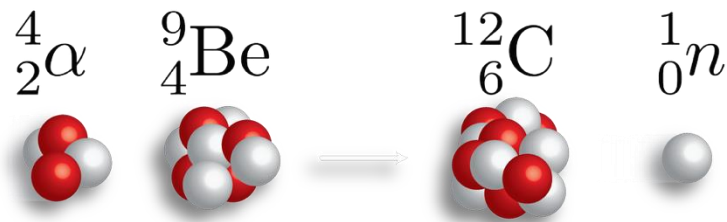
- Energi-udgangspunktet i såvel reaktorer og spallationskilder er "lige højt nok"
- - Målet er bølgelængder i Å-området / energier i meV (**lille** m, ikke store M)



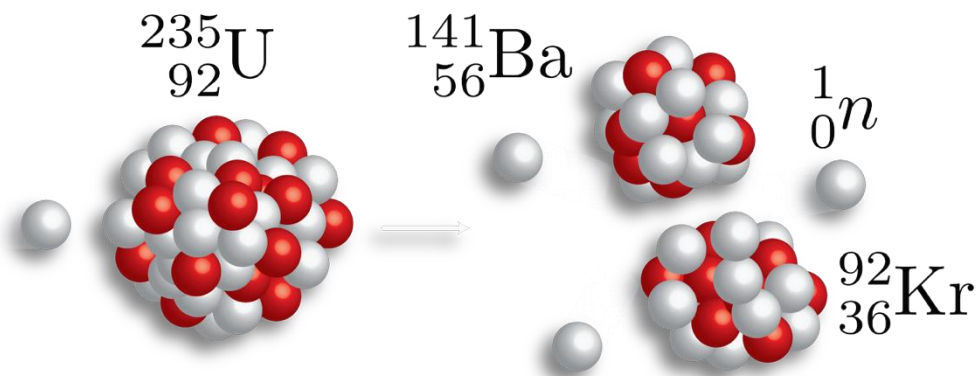
- I neutronspredning taler vi om *termiske* og *kolde* neutroner, hhv. svarende til (f.eks.) medierne H_2O ved stuetemperatur og H(I) eller $\text{CH}_4 \sim 20\text{K}$
- Populært *køler* (modererer) vi neutronerne til passende energi ved "random walk" i "en spand kold vand" eller "flaske flydende brint".
- Resultat: Neutroner med ønskede egenskaber stråles ud i $\sim 4\pi$ rumvinkel

At producere neutroner...

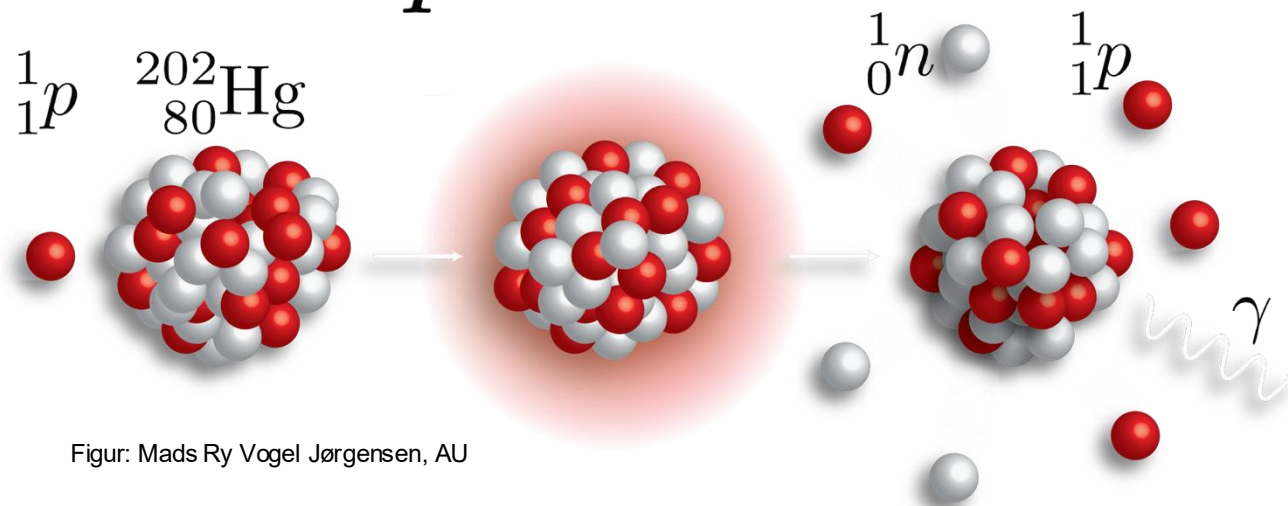
Chadwick



Fission



Spallation



Figur: Mads Ry Vogel Jørgensen, AU

Nuclear process	Example	Neutron yield	Heat release (MeV/n)
d-t in solid target	400 keV d on t in Ti	4×10^{-5} n/d	10,000
Deuteron stripping	40 MeV d on liquid Li	7×10^{-2} n/d	3500
Nuclear photo effect from e^- bremsstrahlung	100 MeV e^- on ${}^{238}\text{U}$	5×10^{-2} n/ e^-	2000
${}^9\text{Be}(d, n){}^{10}\text{Be}$	15 MeV d on Be	1 n/d	1000
${}^9\text{Be}(p, n; p, pn)$	11 MeV p on Be	5×10^{-3} n/p	2000
Nuclear fission	Fission of ${}^{235}\text{U}$ by thermal neutrons	1n/fission	180
Spallation	800 MeV p on ${}^{238}\text{U}$ or Pb	27 n/p or 17 n/p	55 or 30

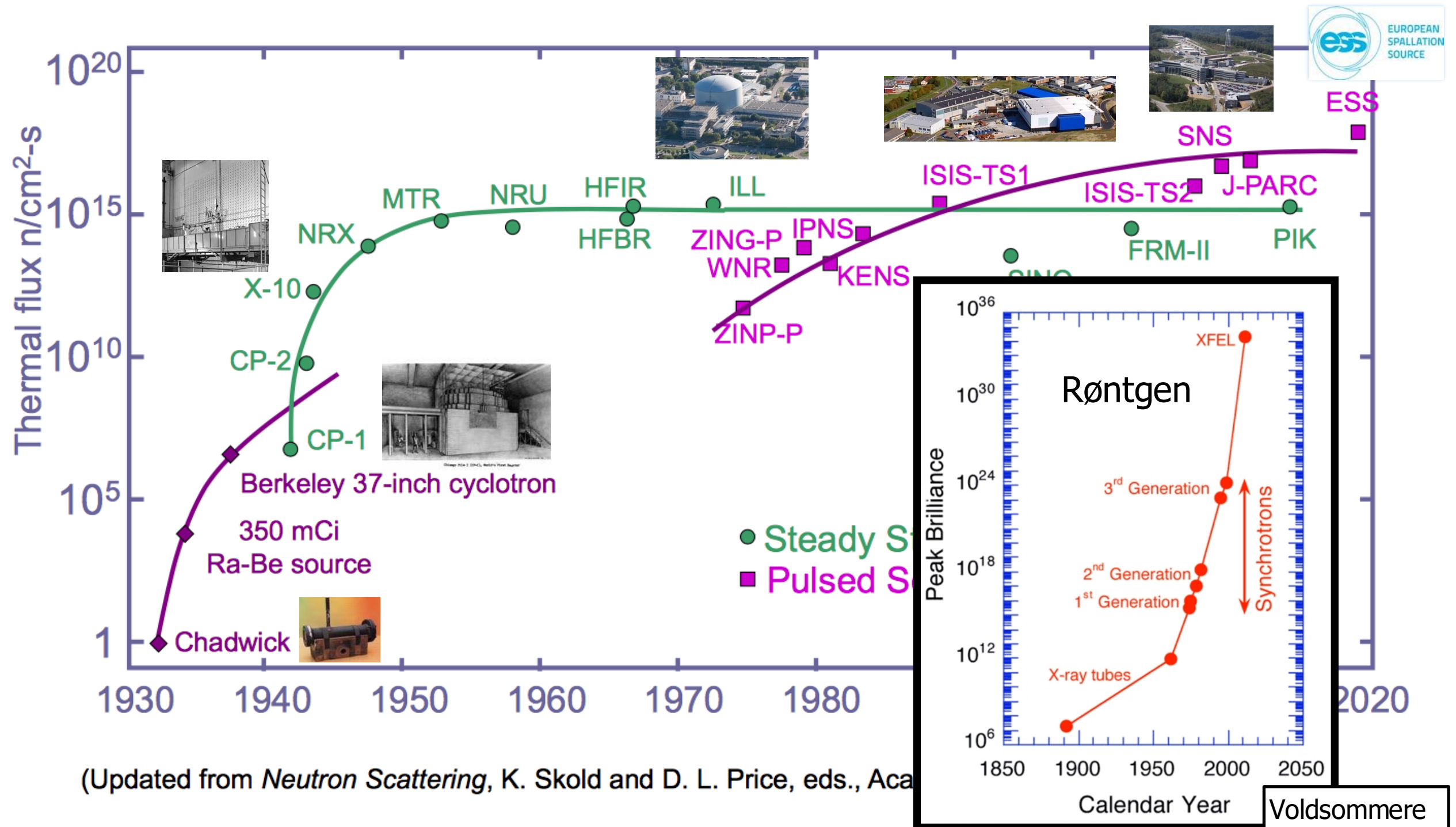
Accelerators for Neutron Generation and Their Applications

Reviews of Accelerator Science and Technology
 Vol. 4 (2011) 219–233
 © World Scientific Publishing Company
 DOI: [10.1142/S1793626811000549](https://doi.org/10.1142/S1793626811000549)

*Fremtiden (50år +) kan betyde brug af fusions-reaktorer, men de bruges pt. til andre formål.... ;-)

Performance af neutronkilder

- I enheder af "brugbare neutroner", dvs. instantan brillians (det kommer vi tilbage til hvad er...)

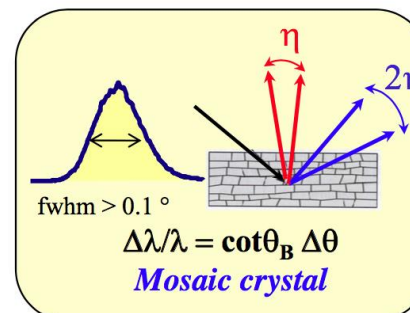
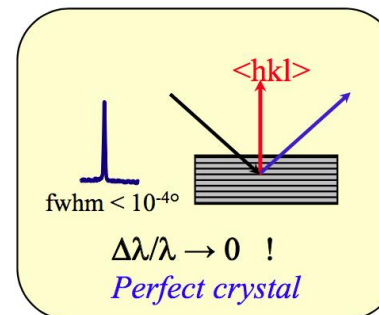
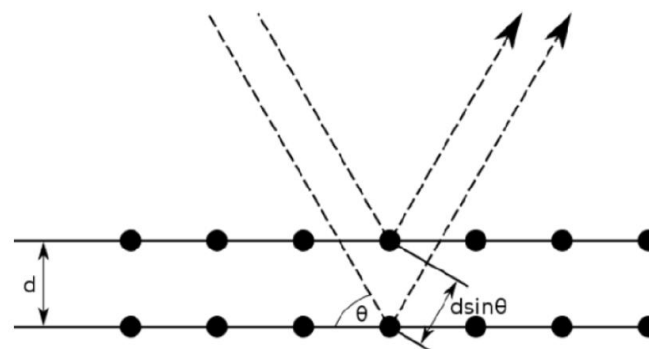


Voldsommere
fremskrivning i
X-ray land....

Reaktor vs. spallationskilde: *konstant vs. pulset (typisk)*

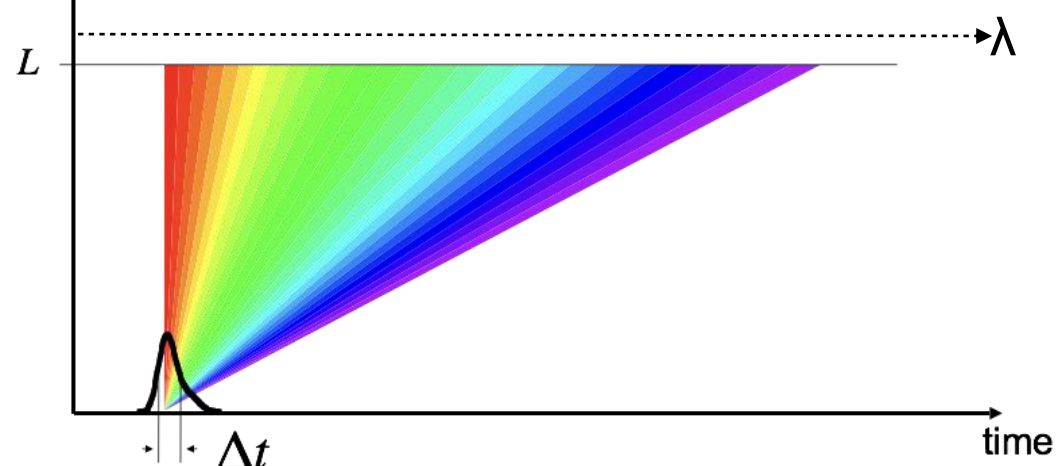
- Jfr. (f.eks.) Braggs lov har vi behov for at vide "hvad bølgelængden er"
- På reaktorer benytter vi typisk netop Braggs lov til at vælge denne bølgelængde:

Bragg's law:
 $\lambda = 2d\sin\theta$



distance

- På pulsed kilder benytter vi istedet Time-of-Flight (ToF)



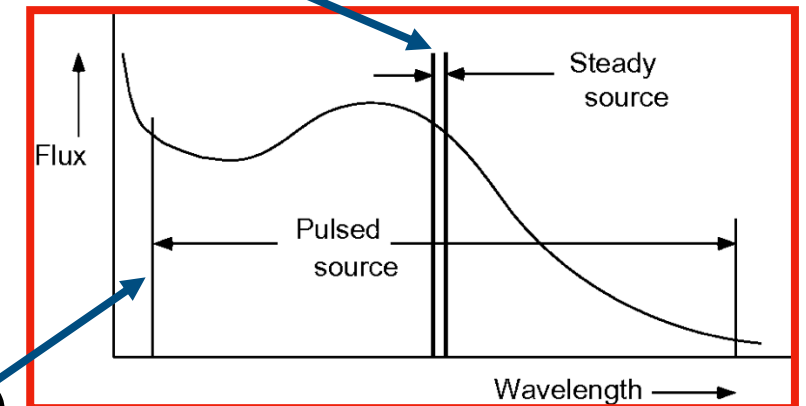
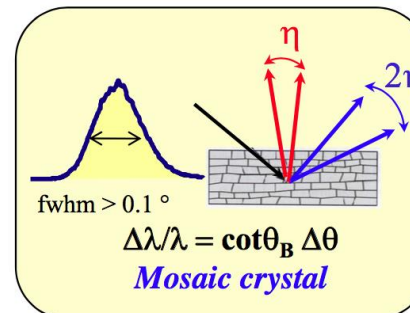
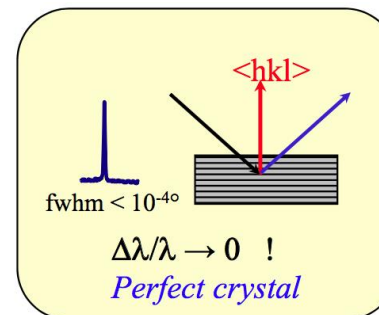
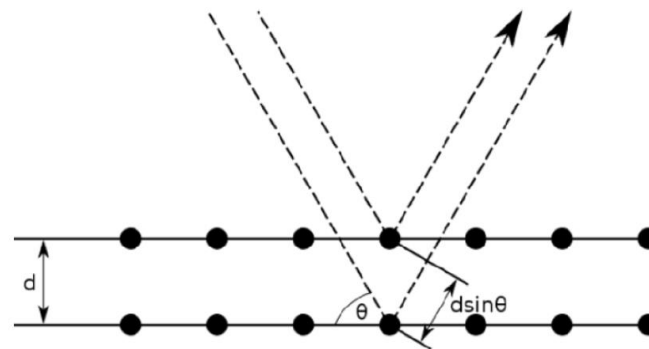
Reaktor vs. spallationskilde:

konstant vs. pulset (typisk)

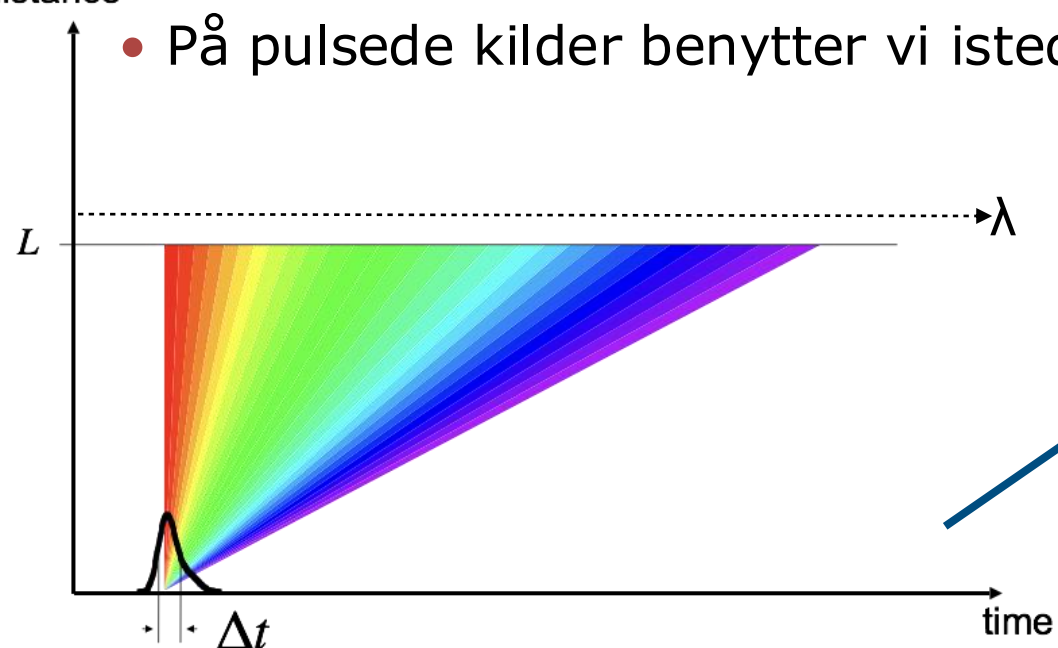
- Jfr. (f.eks.) Braggs lov har vi behov for at vide "hvad bølgelængden er" når vi måler "afbøjet stråling"
- På reaktorer benytter vi typisk netop Braggs lov til at vælge denne bølgelængde:

Bragg's law:

$$\lambda = 2d\sin\theta$$



distance

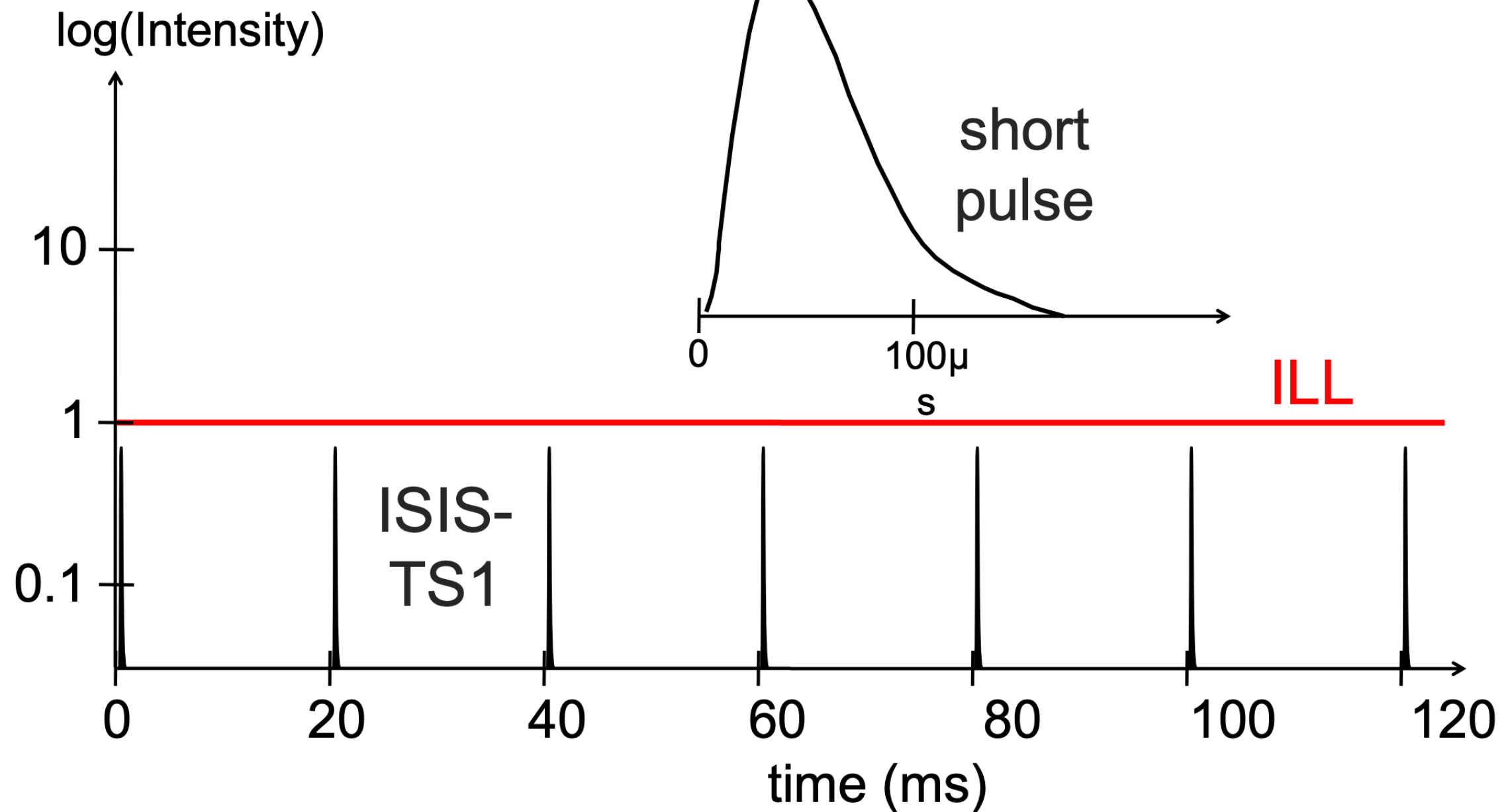


- På pulsed kilder benytter vi istedet Time-of-Flight (ToF)

ILL vs. de pulsedde kilder - performance



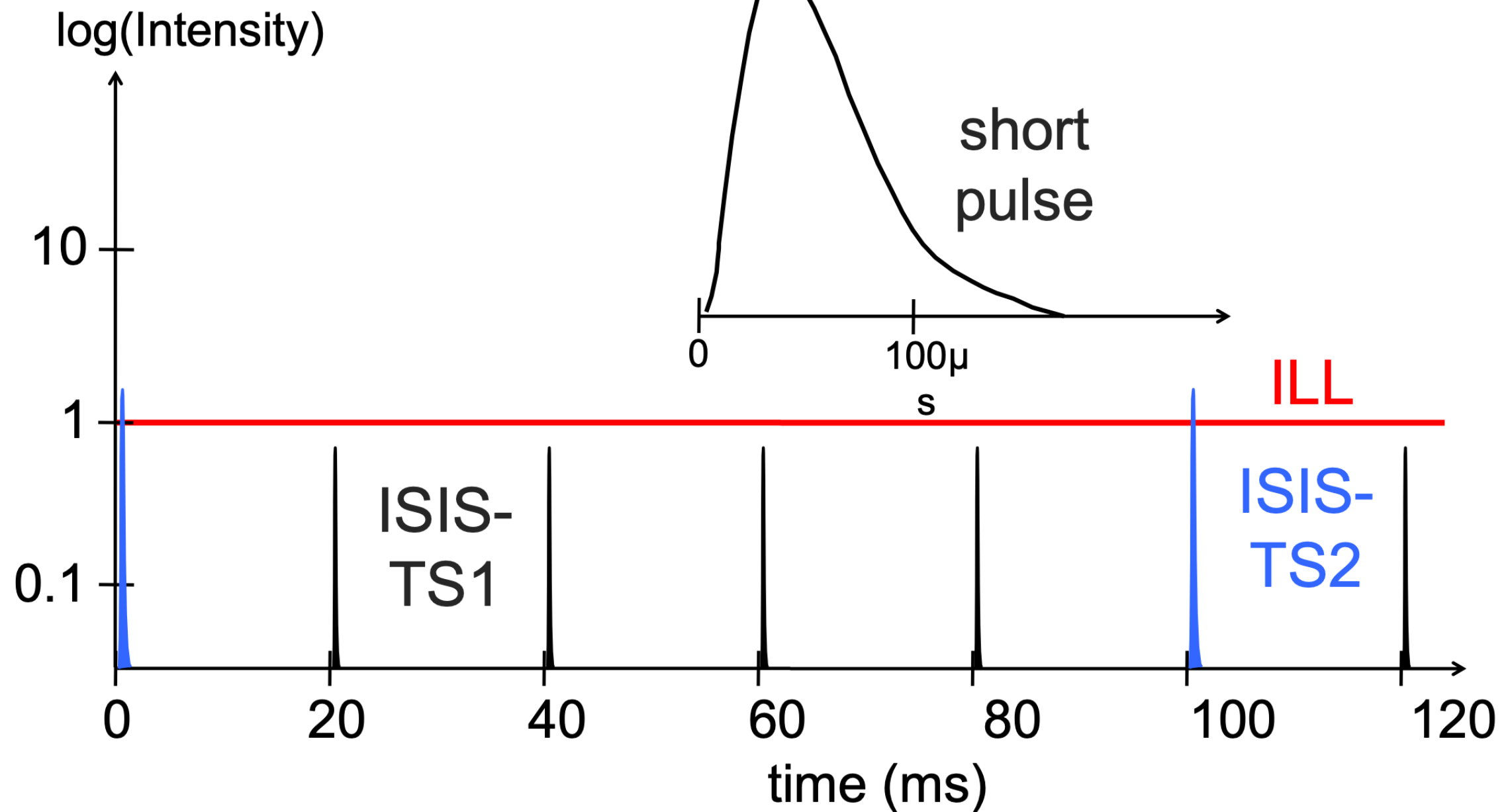
Pulsed-source time structures cold neutrons



ILL vs. de pulsedde kilder - performance



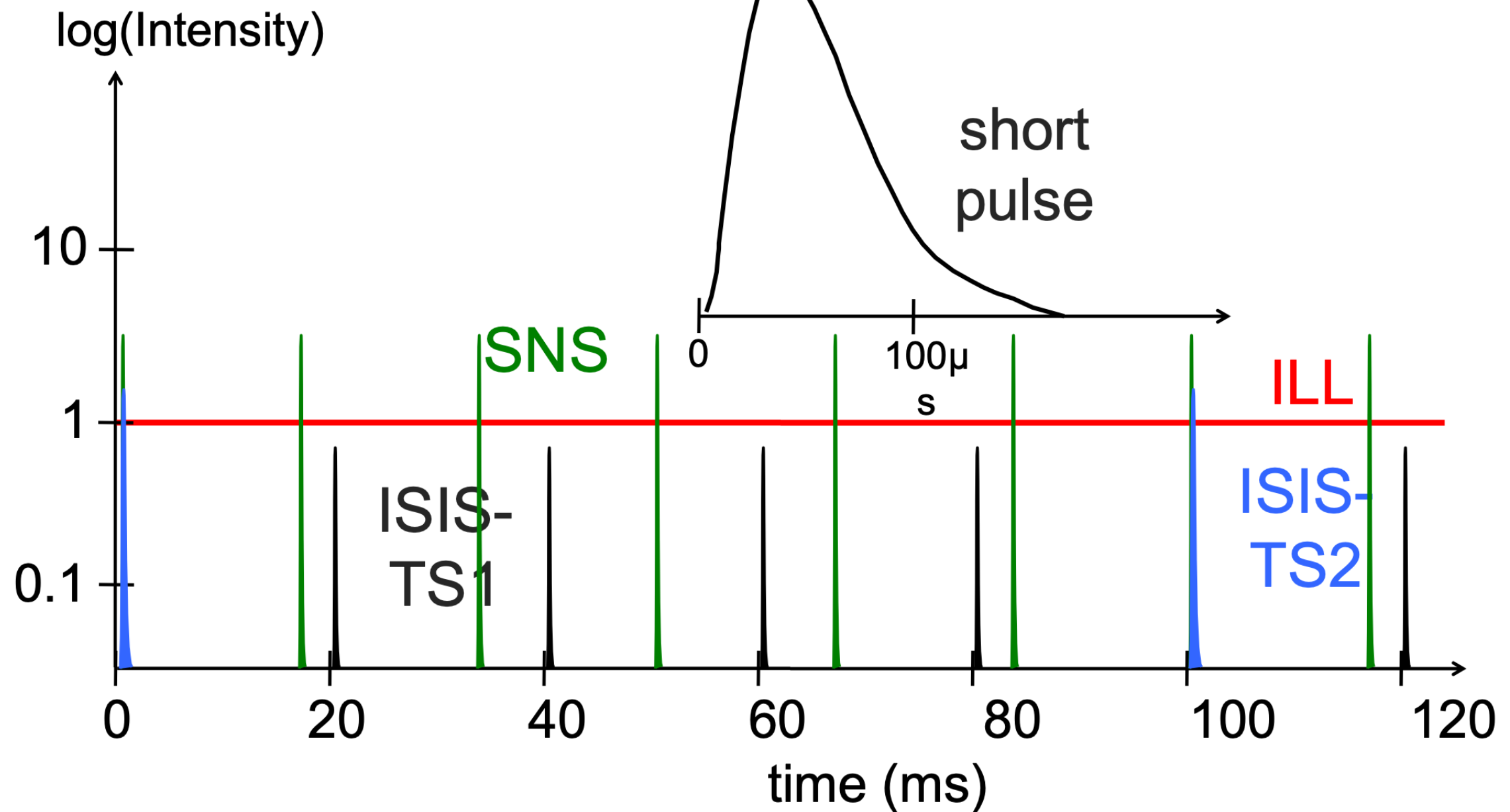
Pulsed-source time structures cold neutrons



ILL vs. de pulsedde kilder - performance



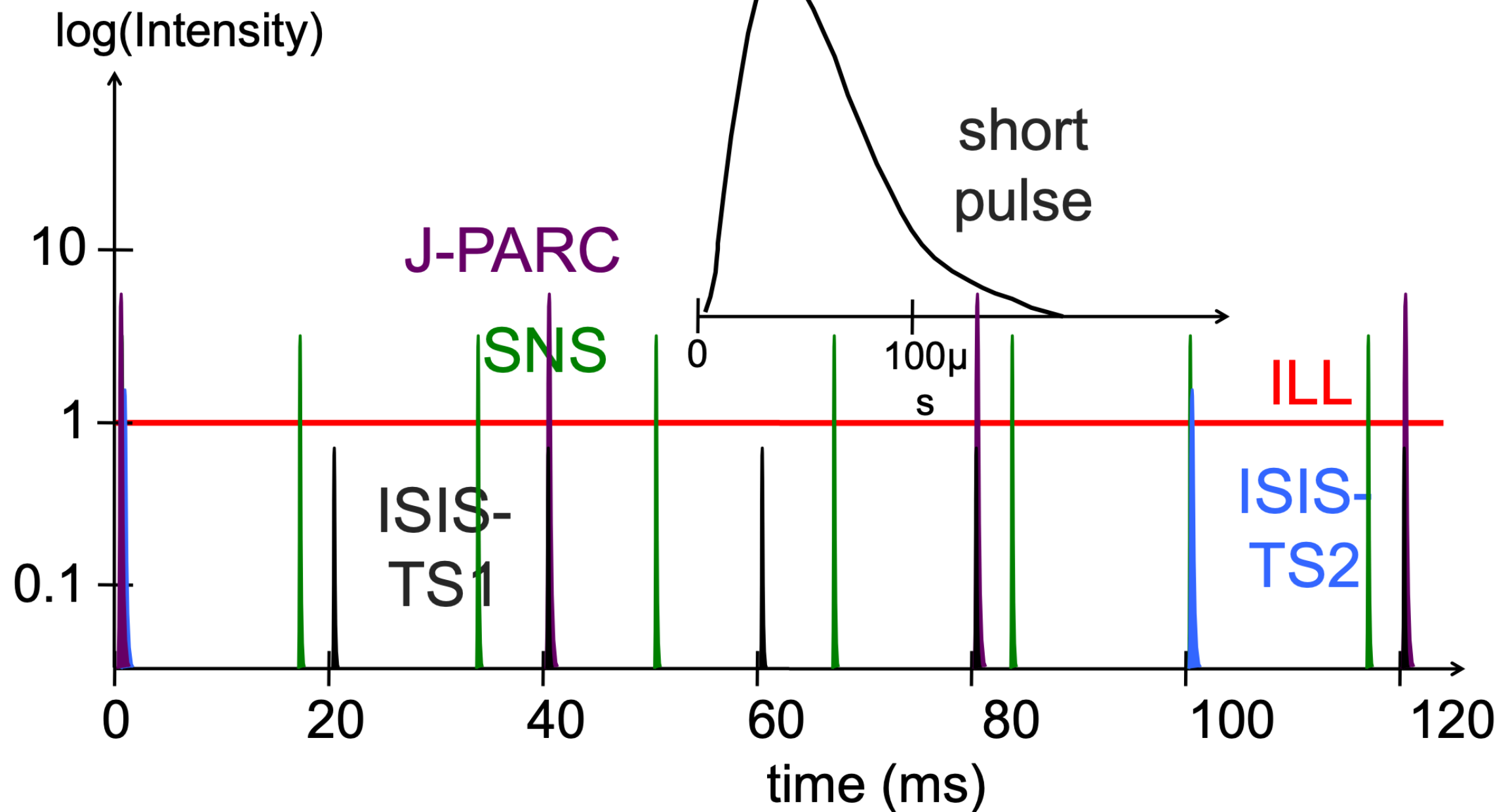
Pulsed-source time structures cold neutrons



ILL vs. de pulsedede kilder - performance



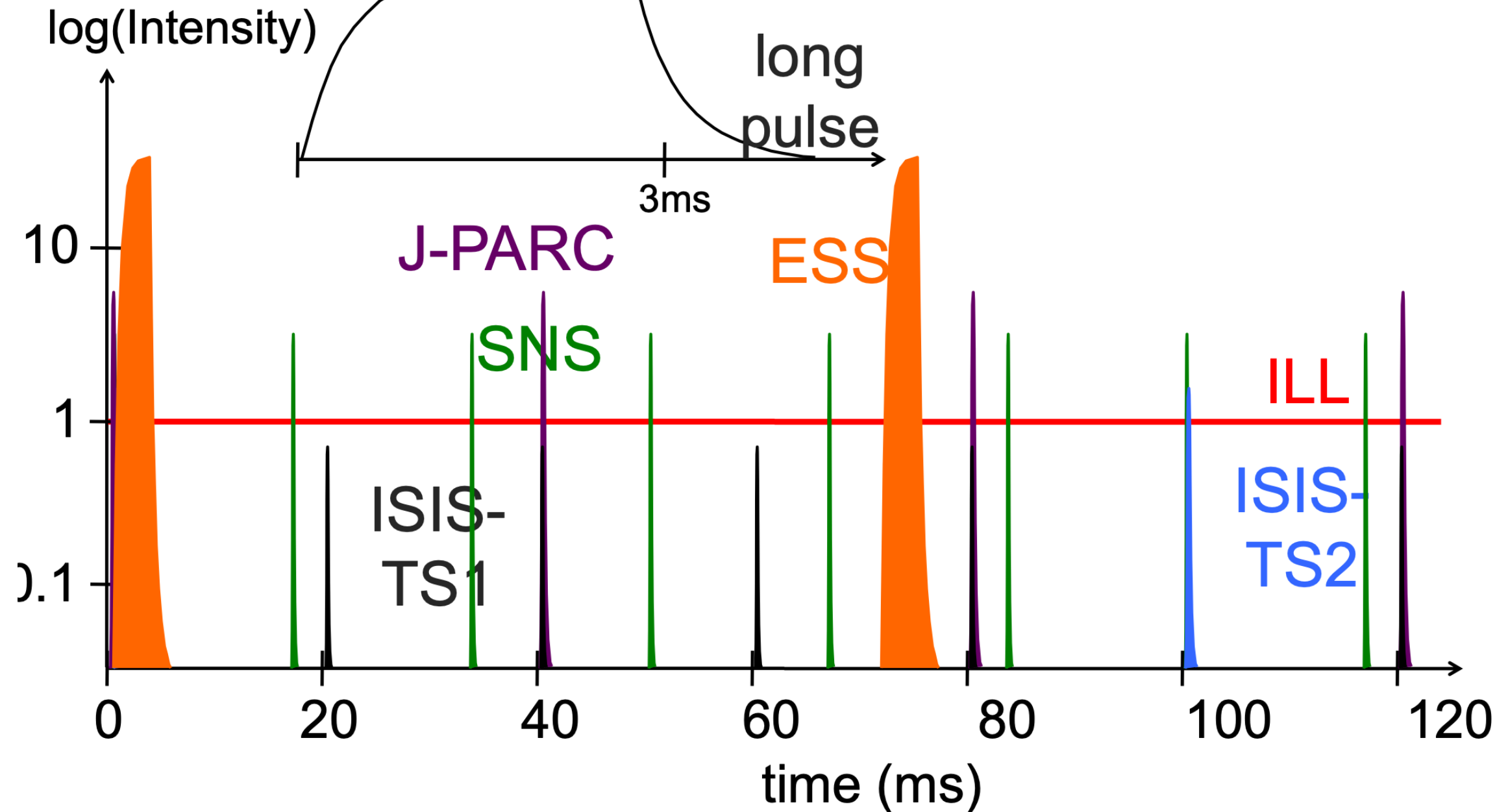
Pulsed-source time structures cold neutrons



ILL vs. de pulsedede kilder - performance



Pulsed-source time structures cold neutrons

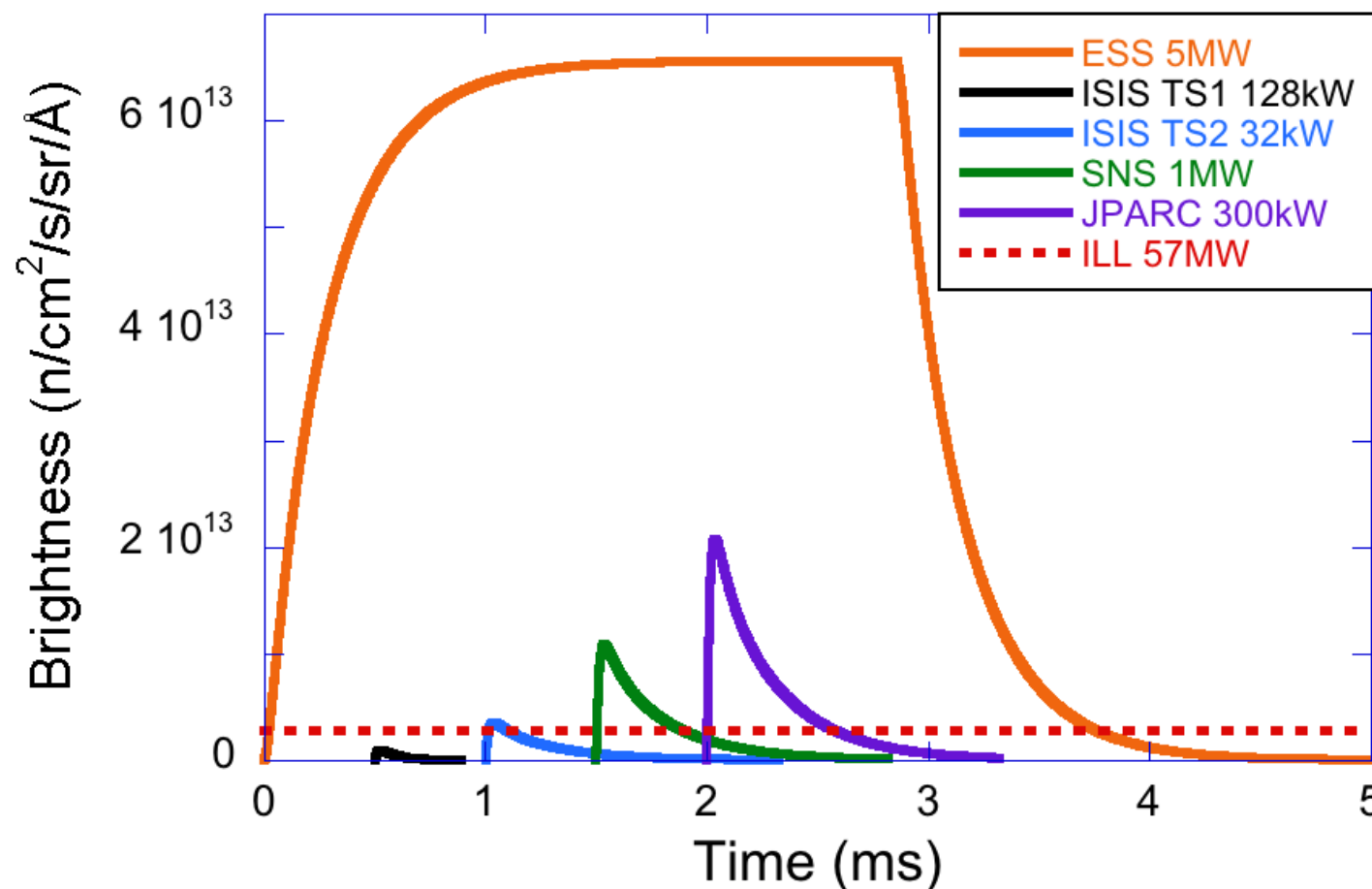


Altså...



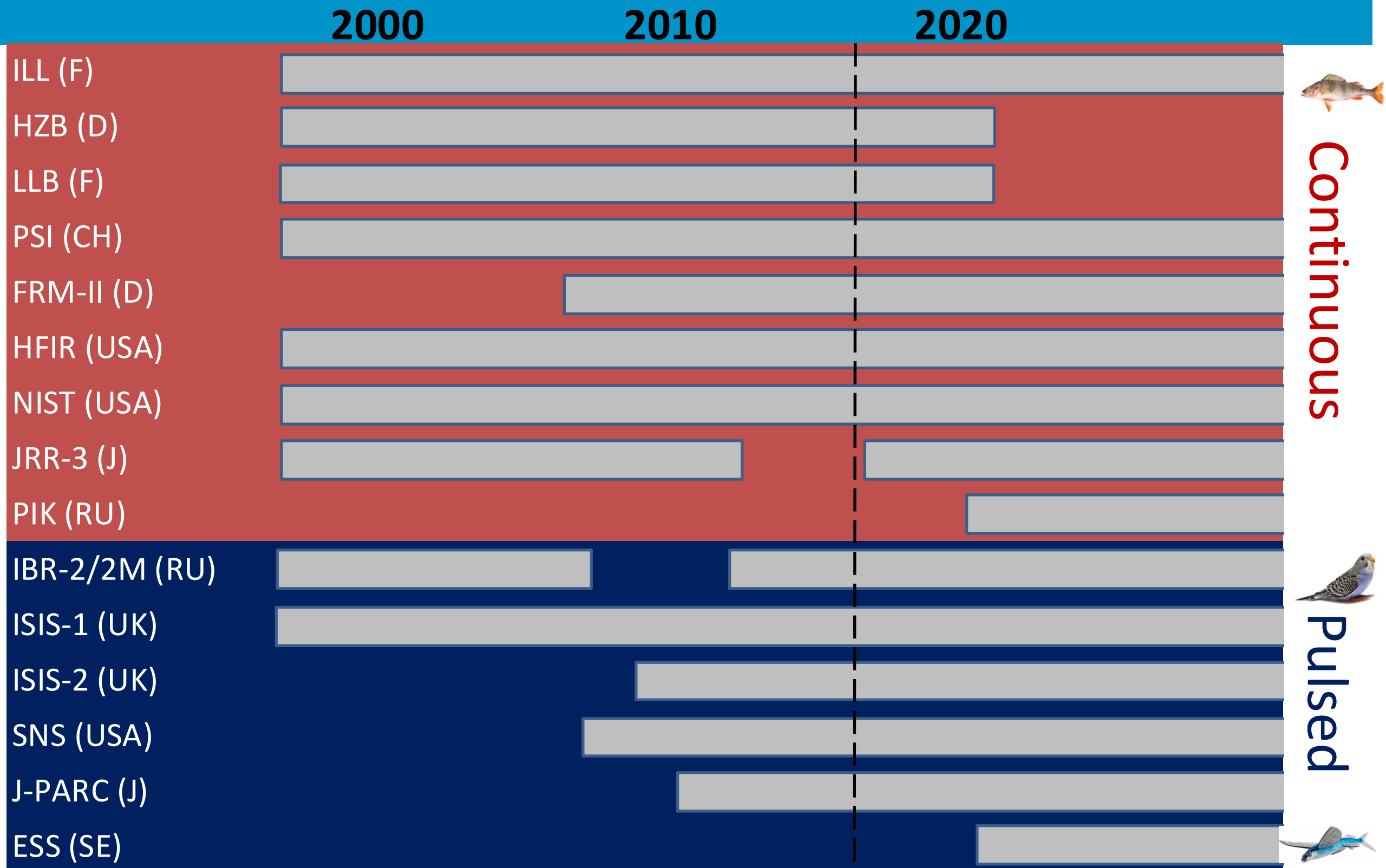
- ESS er hverken en steady-state kilde eller en kortpulset kilde, men langpulset

Single-Pulse Brightness at 5 Å

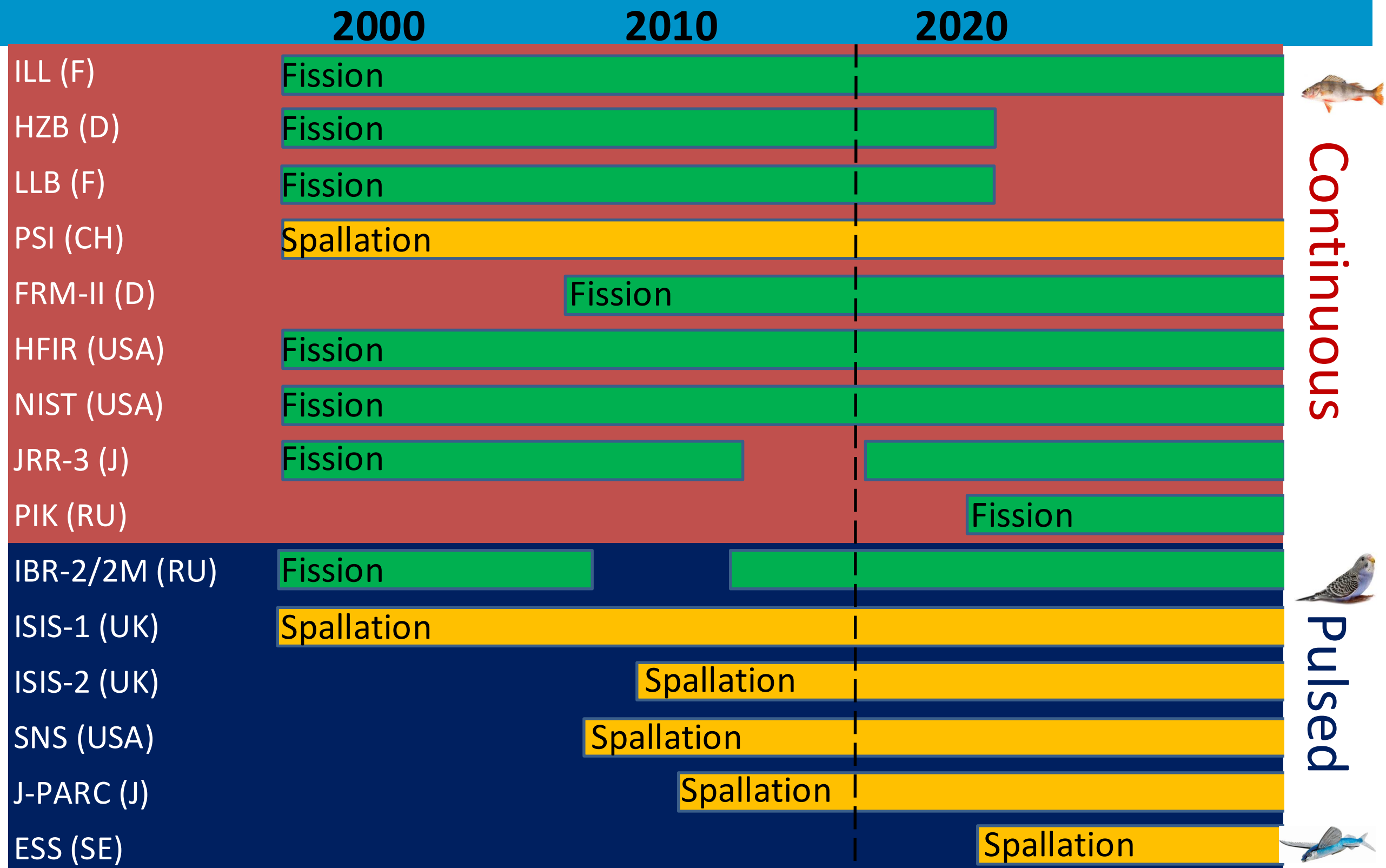


- Med *passende instrumentering* der tager højde for dette kan man opnå (mindst) 30 x ILL performance og 5-10 x kortpulset performance

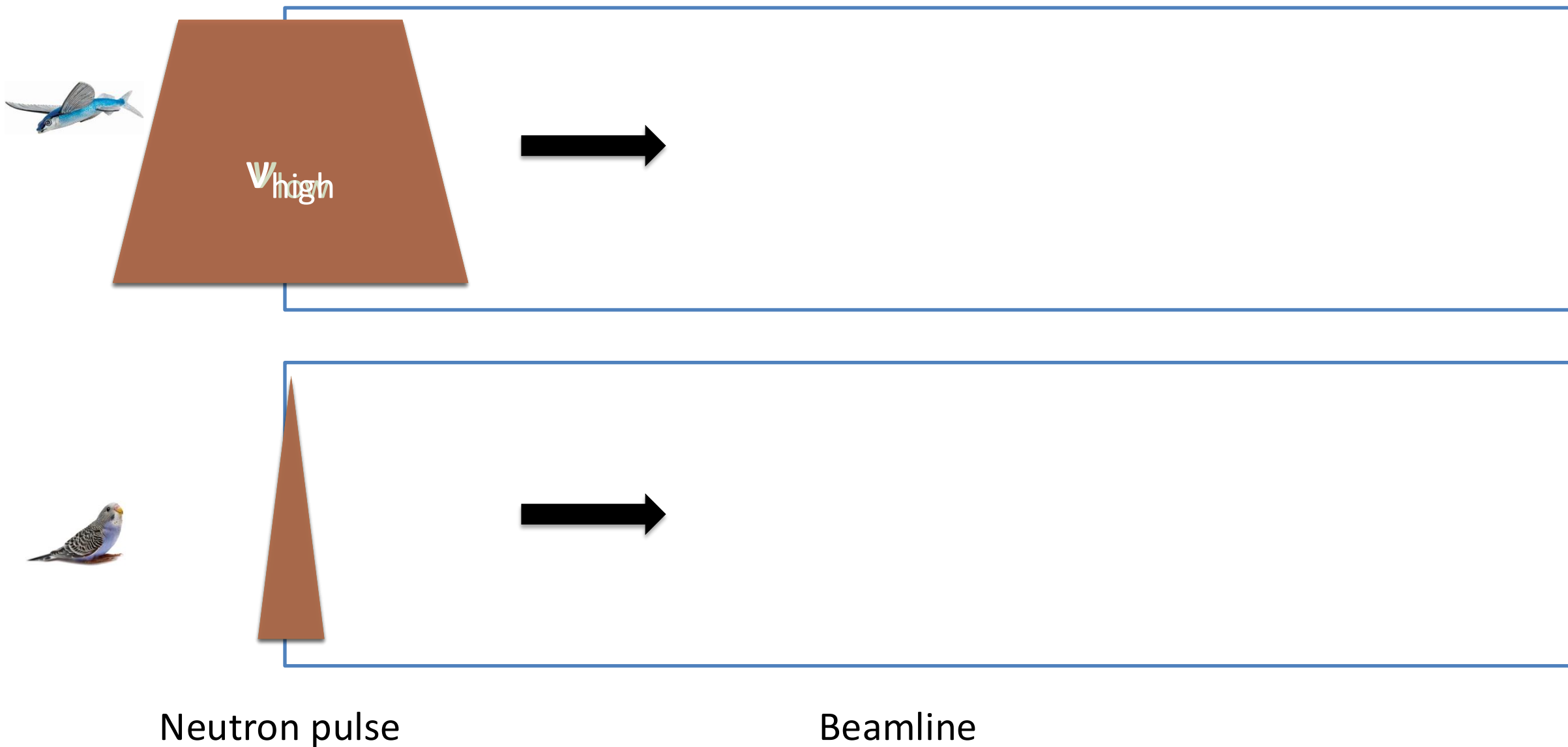
Major neutron sources in the world



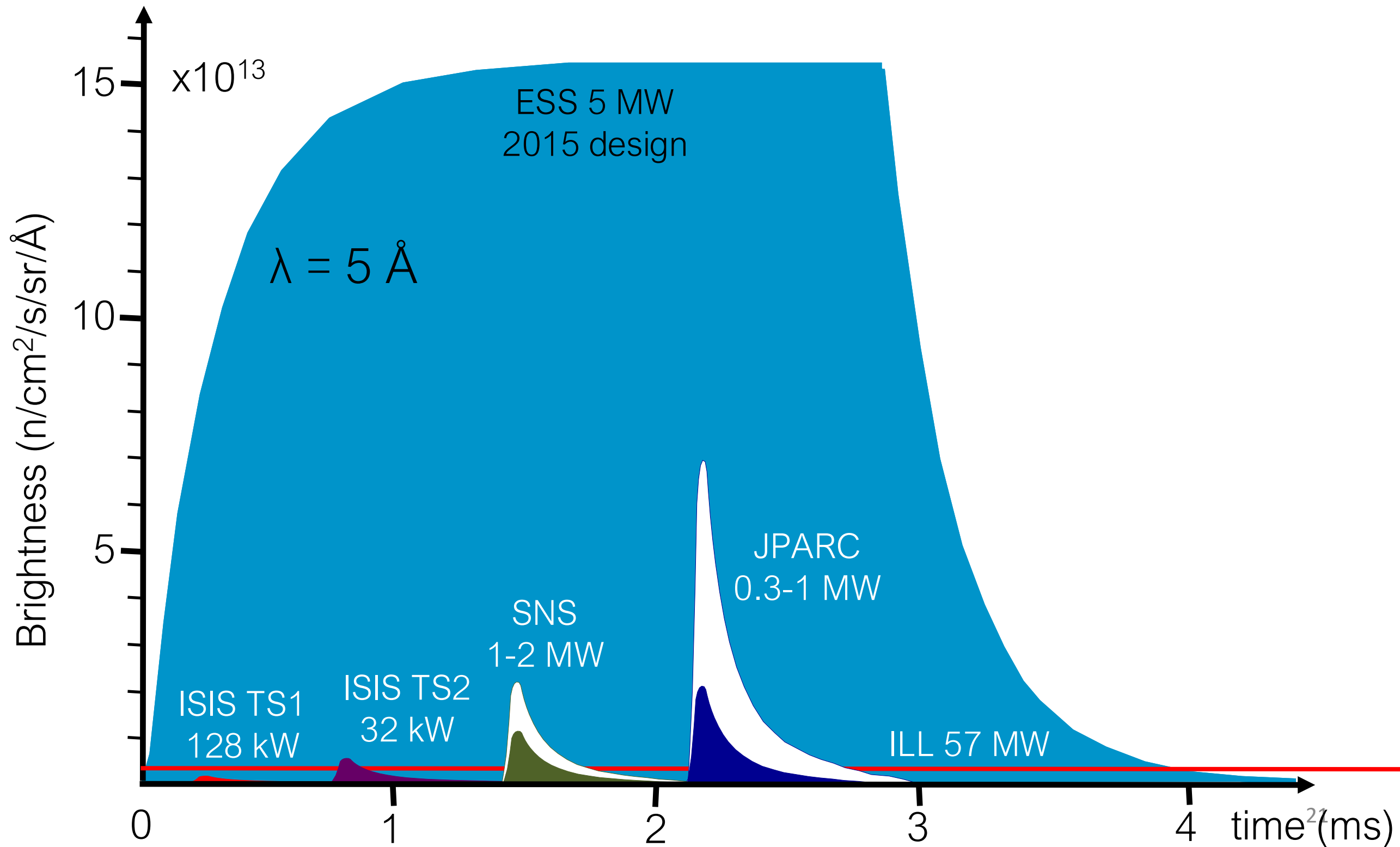
Major neutron sources in the world



Lengths of beam line and pulse affect λ resolution



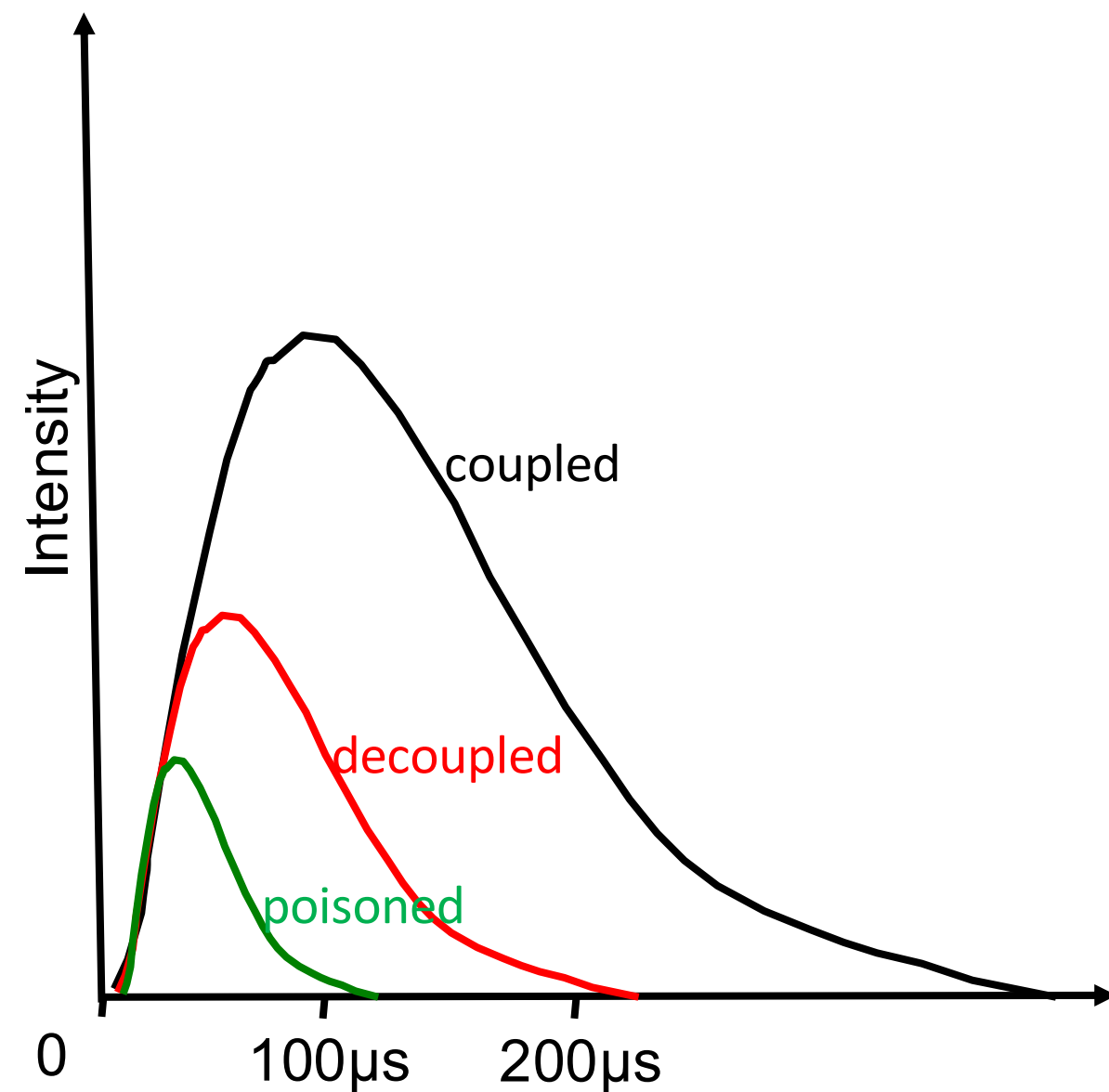
Long-pulse performance



Adapting the pulse width

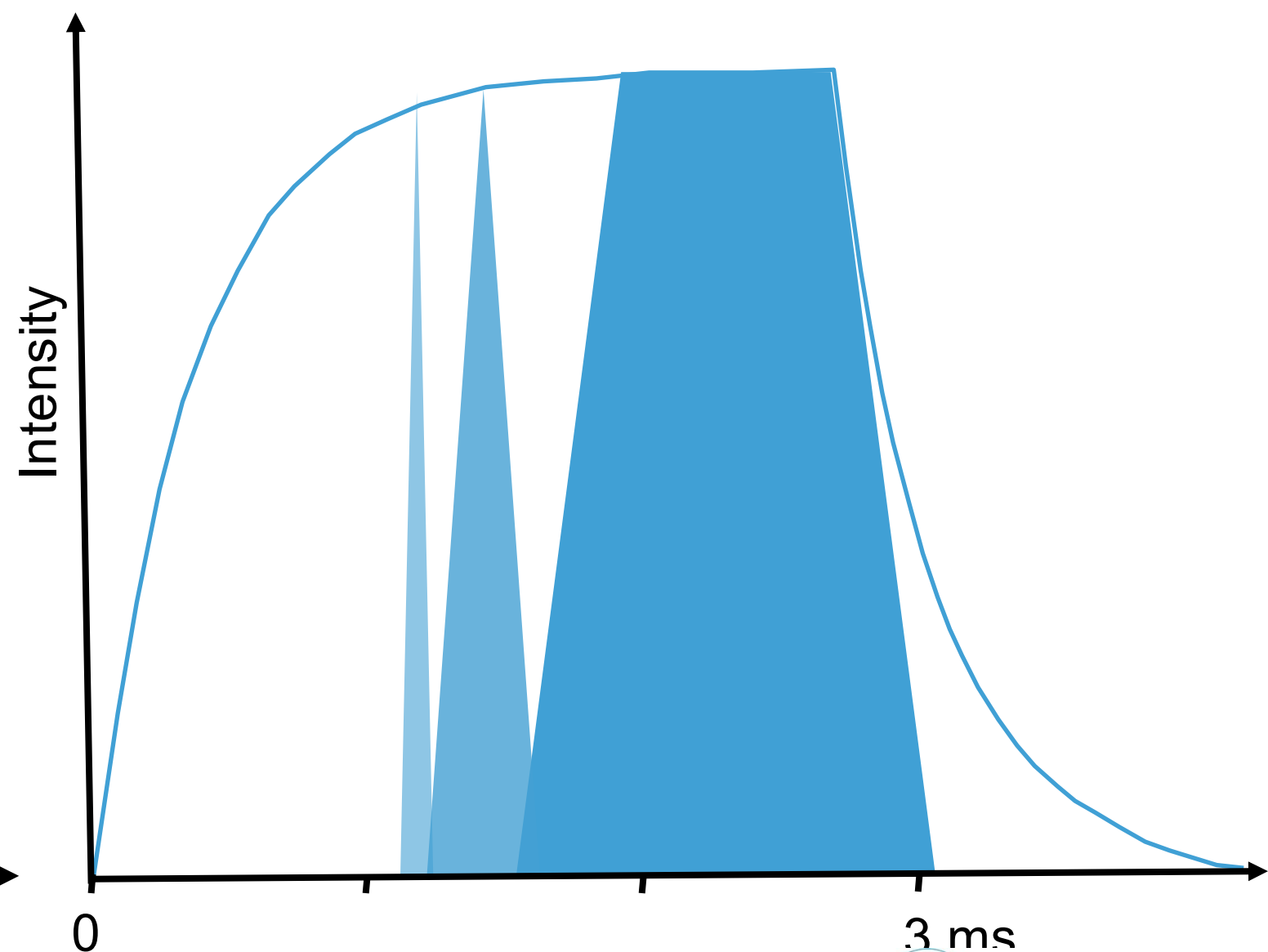
Short-Pulse Source

- set pulse width by choosing moderator

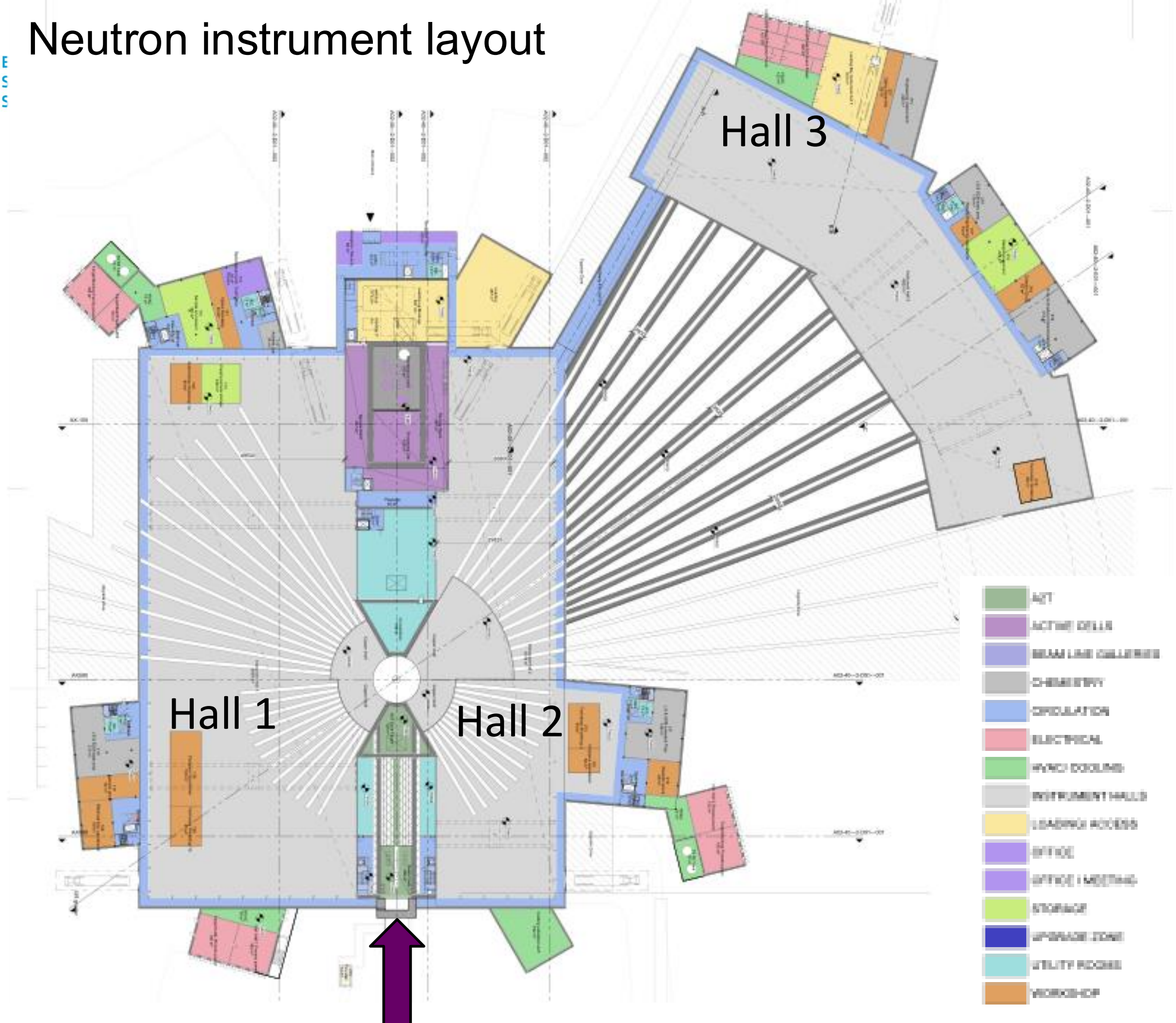


Long-Pulse Source (ESS)

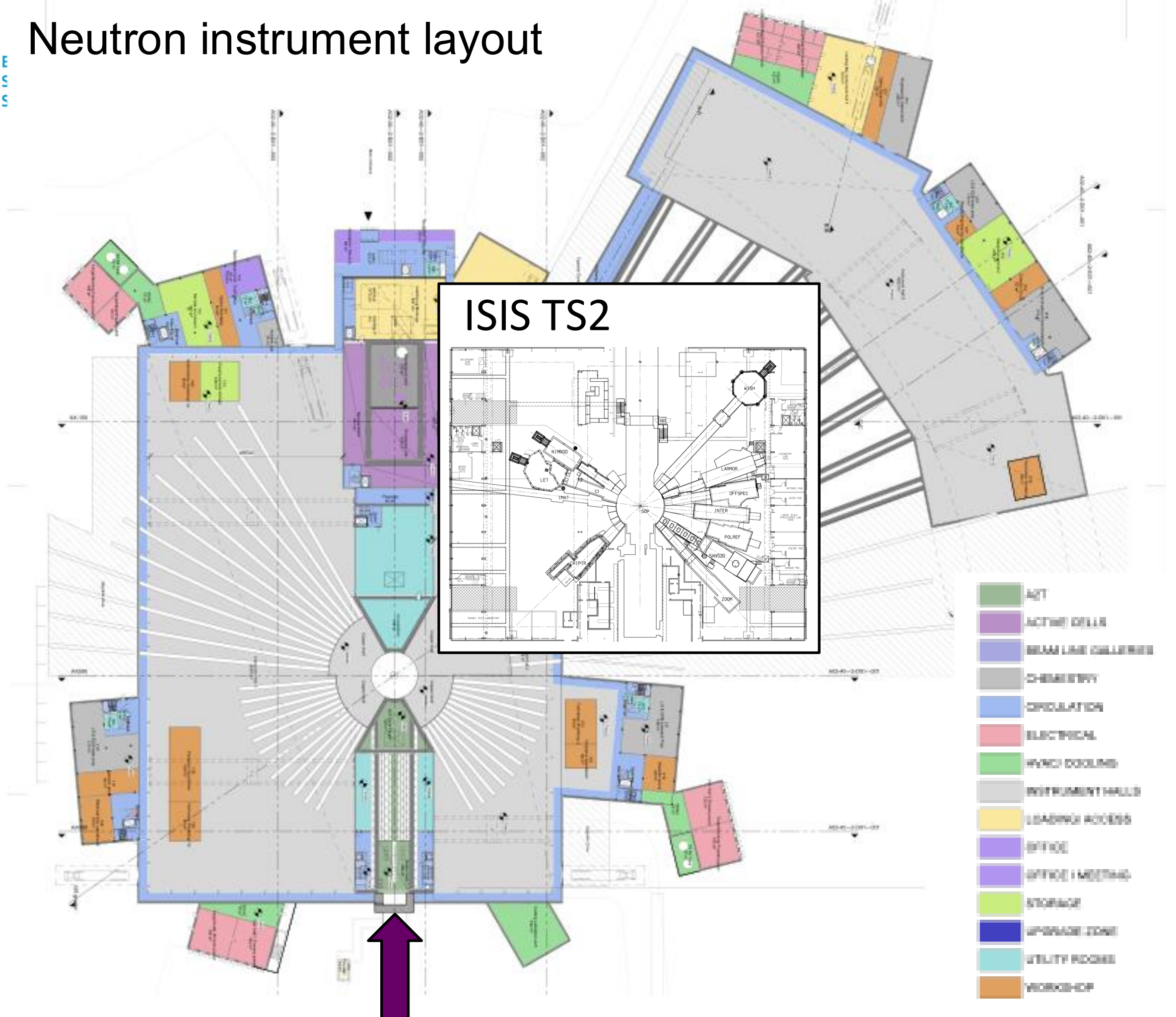
- set pulse width using pulse-shaping chopper



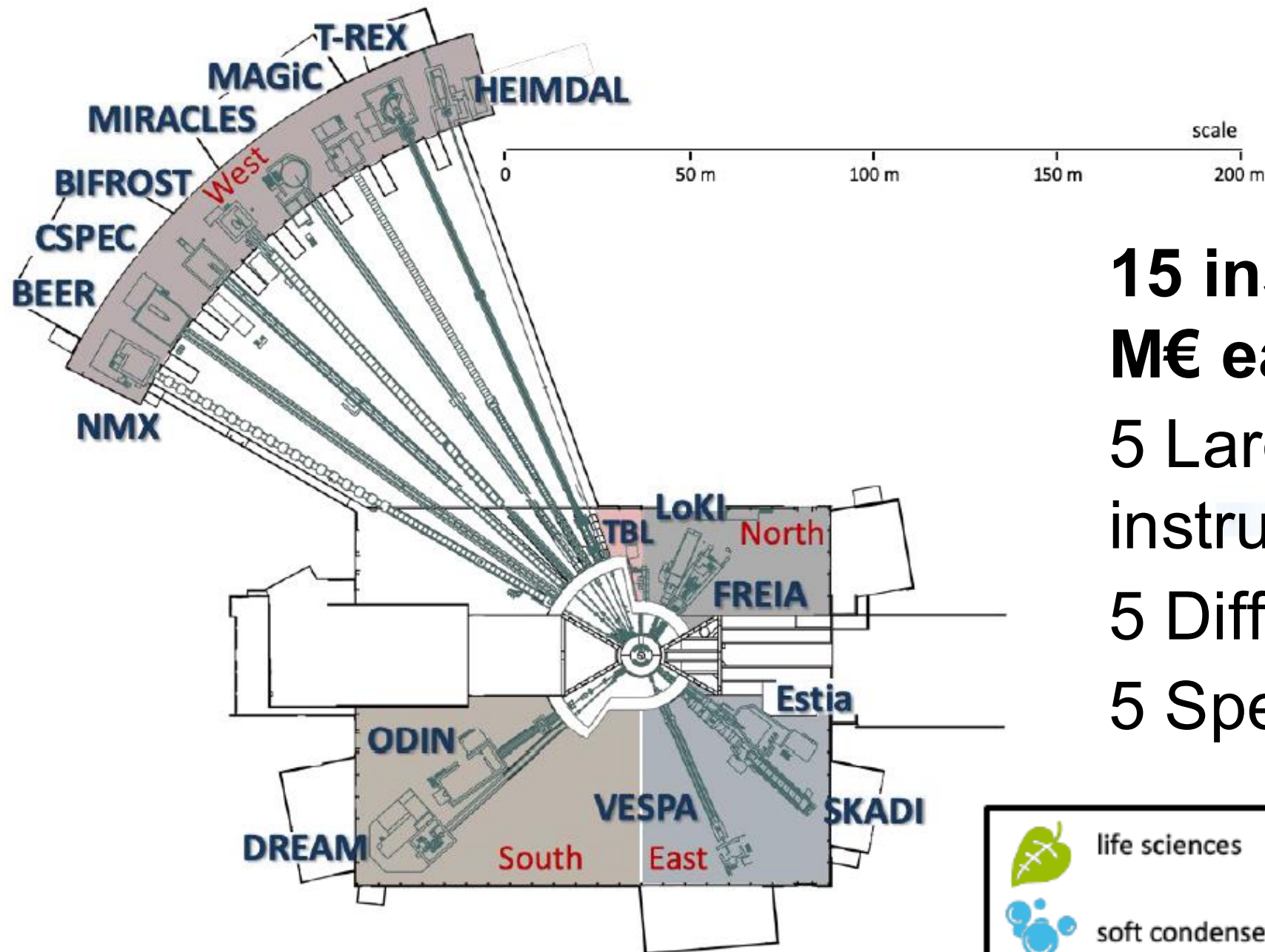
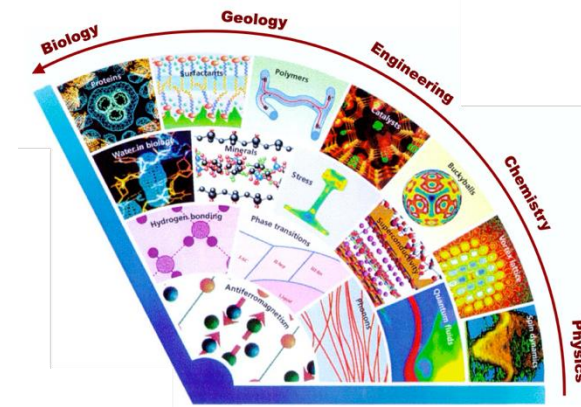
Neutron instrument layout



Neutron instrument layout



Instrument suite at ESS

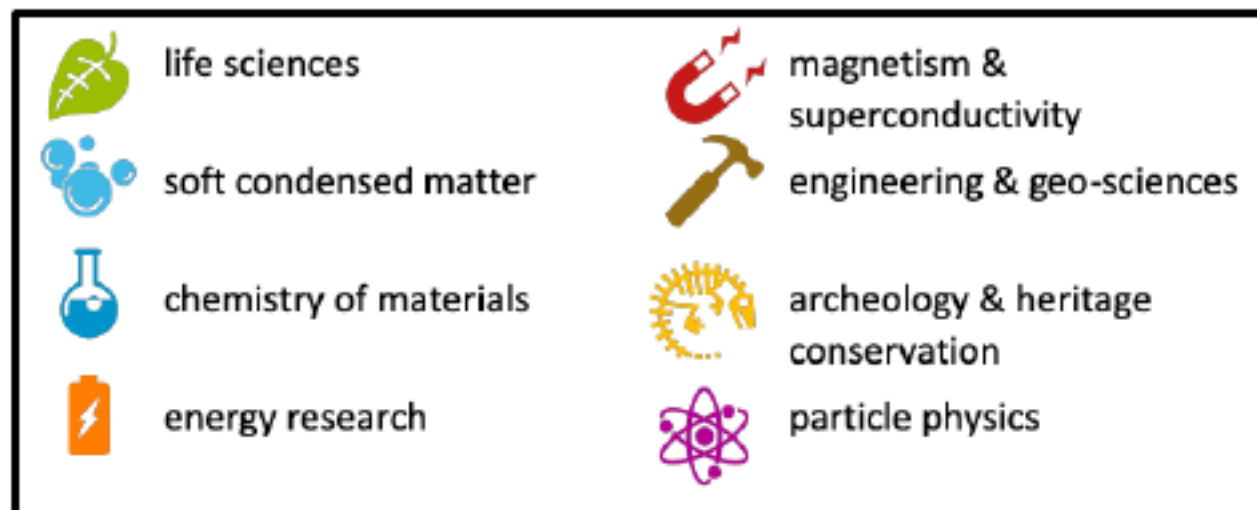


**15 instruments (10-15
M€ each)**

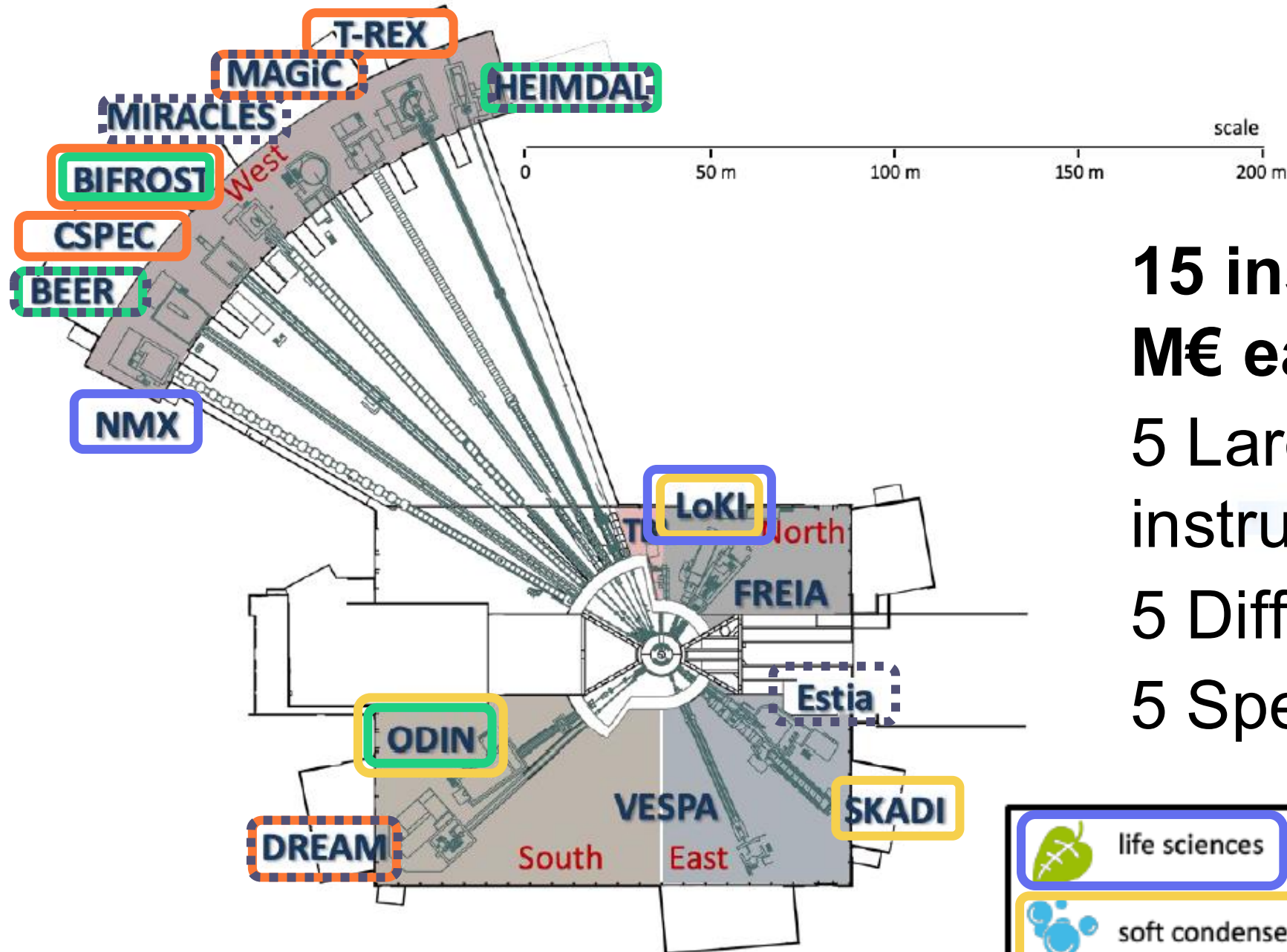
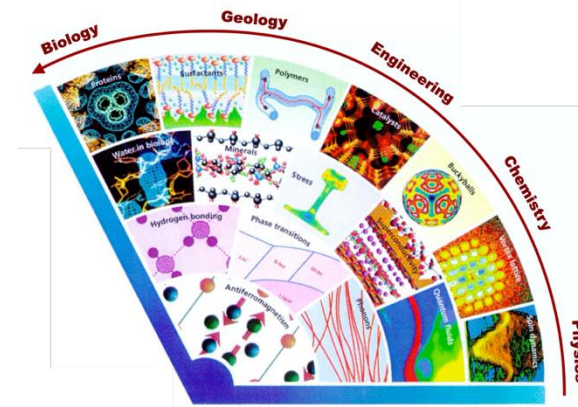
**5 Large-scale structure
instruments**

5 Diffractometers

5 Spectrometer



Instrument suite at ESS



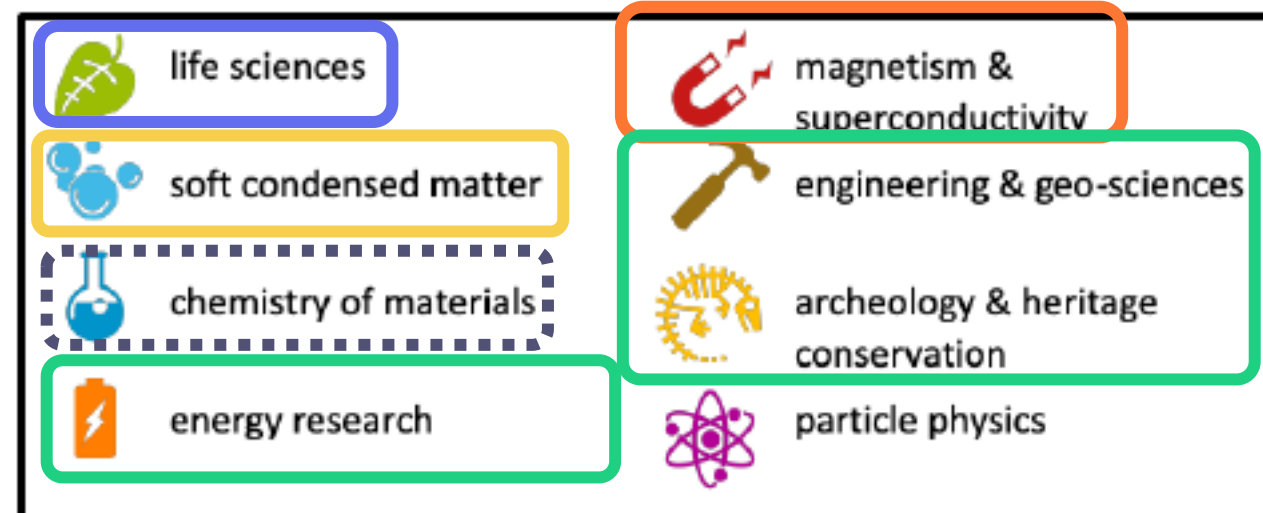
15 instruments (10-15 M€ each)

5 Large-scale structure instruments

5 Diffractometers

5 Spectrometer

*classification not official



Instrument suite at ESS

Large-Scale Structures

ODIN Imaging Instrument	    
SKADI General Purpose SANS	   
LOKI Broadband SANS	 
Surface Scattering	   
FREIA Horizontal Reflectometer	  
Estia Vertical Reflectometer	    
HEIMDAL Powder Diffractometer	   
DREAM Powder Diffractometer	   
Monochromatic Powder Diffractometer	  
BEER Engineering Diffractometer	  
Extreme Conditions Diffractometer	   
MAGiC Magnetism Diffractometer	 
NMX Macromolecular Diffractometer	 

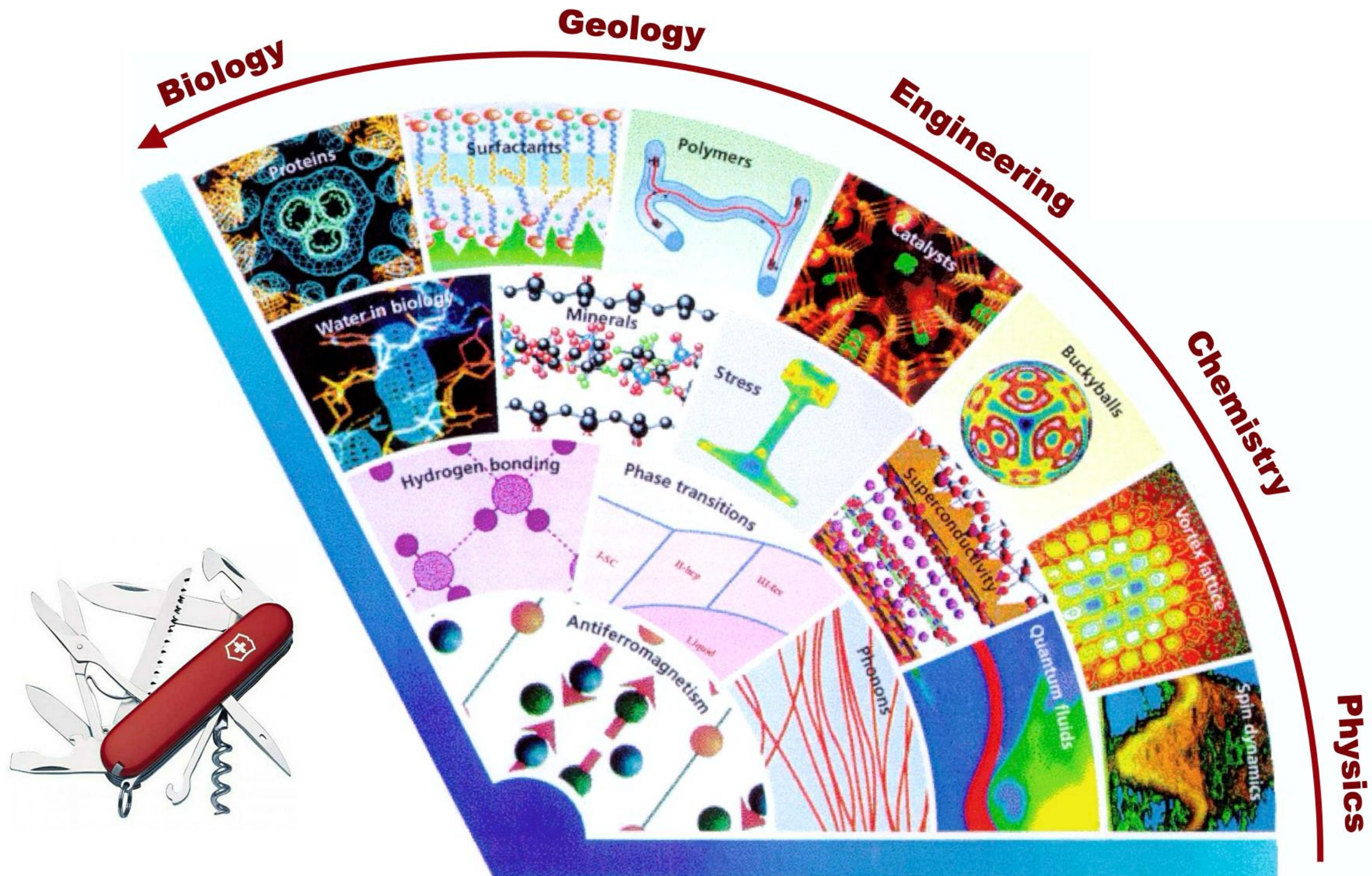
Diffractometer

Spectroscopy

CSPEC Cold Chopper Spectrometer	   
Broadband Spectrometer	   
T-REX Thermal Chopper Spectrometer	  
BIFROST Crystal Analyser Spectrometer	   
VESPA Vibrational Spectroscopy	  
MIRACLES Backscattering Spectrometer	   
High-Resolution Spin-Echo	   
Wide-Angle Spin-Echo	   
Particle Physics Beamline	

 life sciences	 magnetism & superconductivity
 soft condensed matter	 engineering & geo-sciences
 chemistry of materials	 archeology & heritage conservation
 energy research	 particle physics

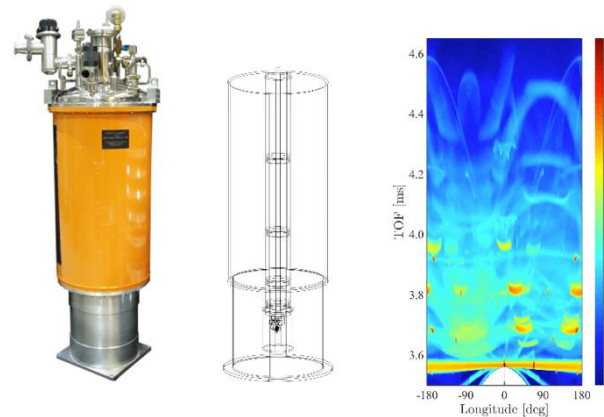
Broad length and time scales



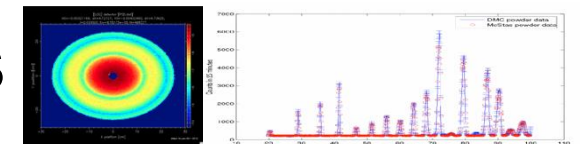
Components of neutron instruments

- At få de rette til det givne e tabsløst som r

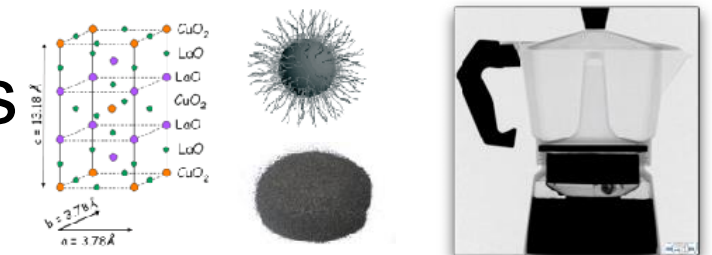
Sample-environments -



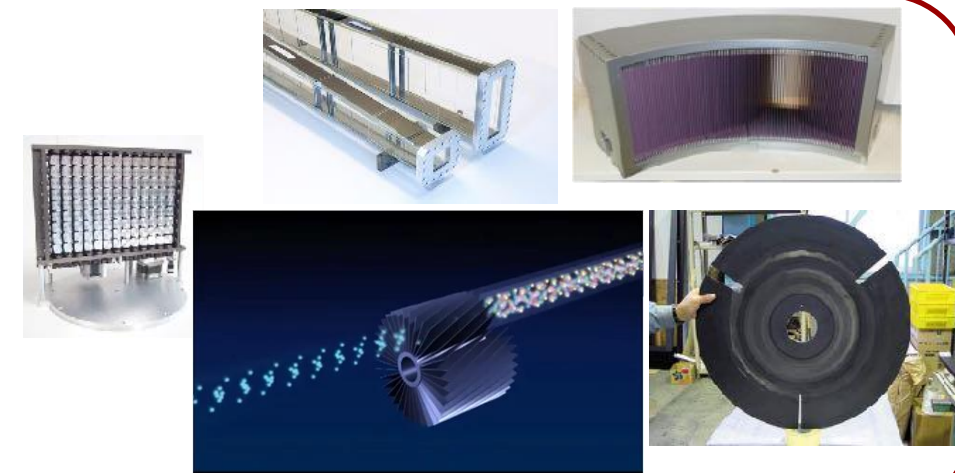
Detectors



Scientific samples



Neutron optics



Neutron moderator

Starts with a source of neutrons, be it a reactor- or spallation source

The European Spallation Source



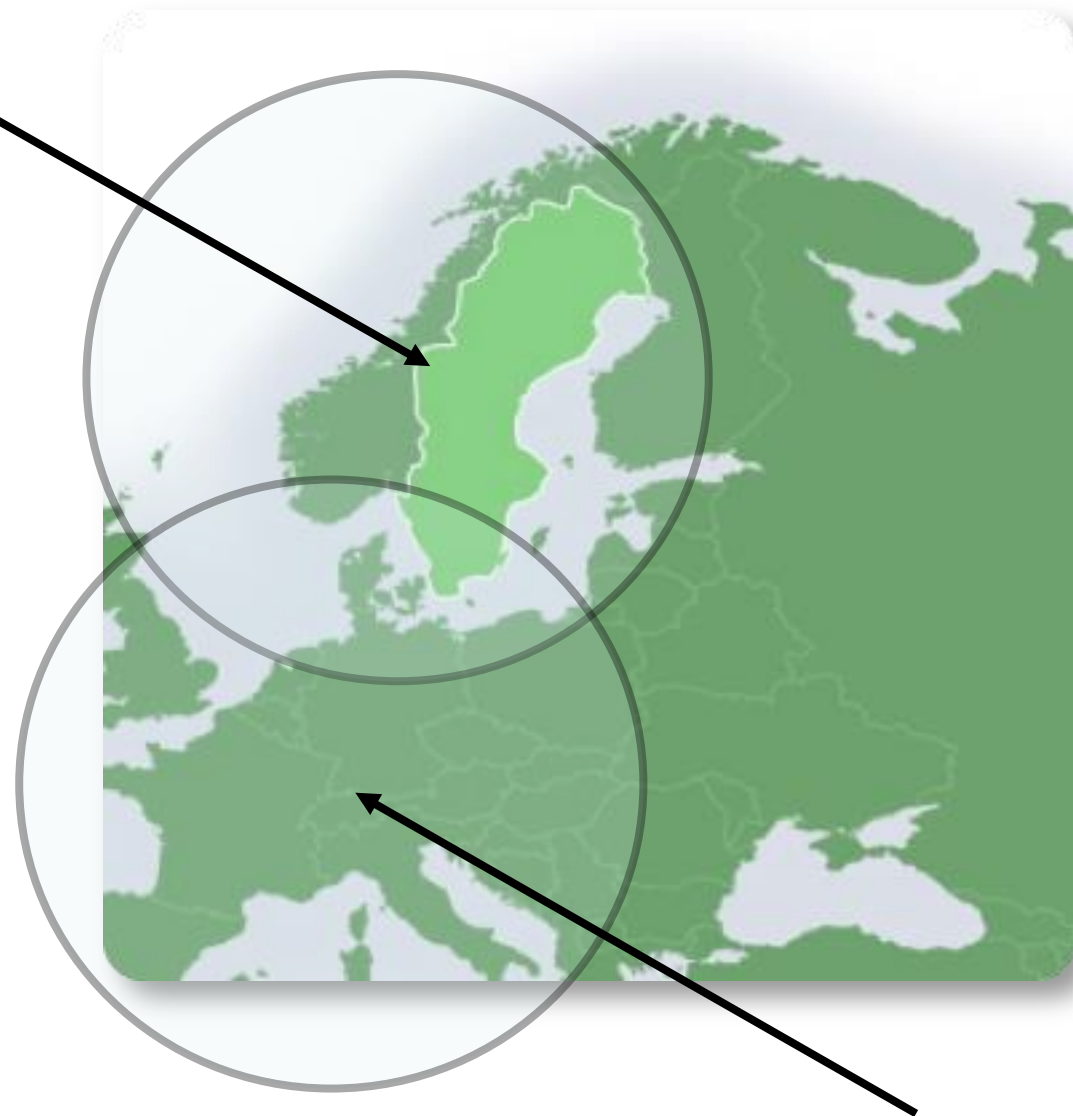
Host countries: Sweden, Denmark

Partner countries: Czech Republic, Estonia, France, Germany, Hungary, Italy, Norway, Poland, Spain, Switzerland and the United Kingdom

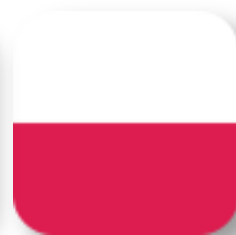


An International Collaboration

Sweden,
Denmark and Norway:
50% of construction and
20% of operations costs



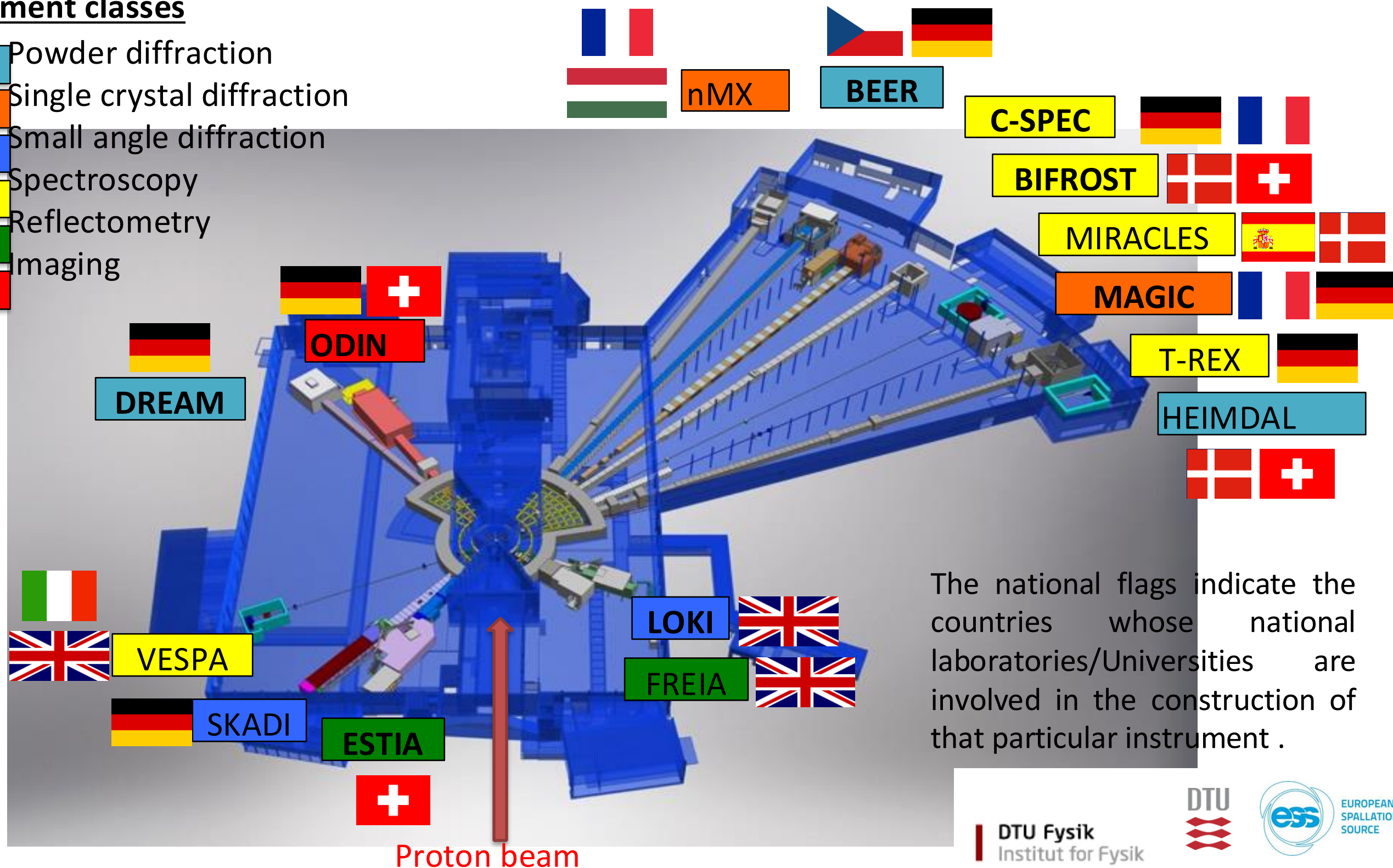
European partners
pays the rest



The ESS Instrument Suite 15/22

Instrument classes

- Powder diffraction
- Single crystal diffraction
- Small angle diffraction
- Spectroscopy
- Reflectometry
- Imaging



The national flags indicate the countries whose national laboratories/Universities are involved in the construction of that particular instrument.

The Data Management and Software Centre

Topic of later, dedicated talk...



- Data Management
- Instrument Control
- Data Reduction & Analysis
- Instrument simulations
- (Theory and simulations)

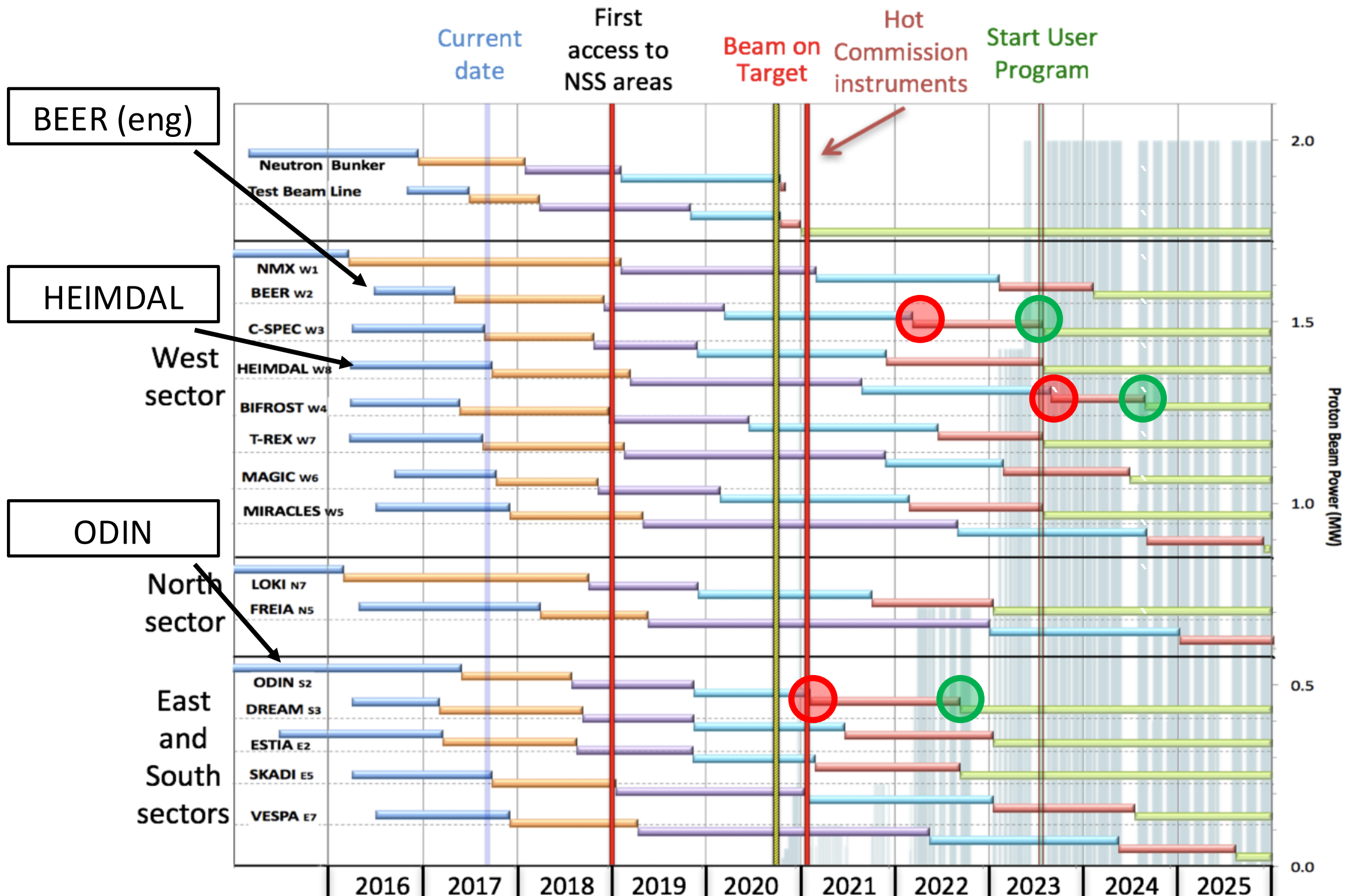


DMSC at DTU in
Lyngby, Denmark



High-level instrument construction schedule

(ældre version – tiden er skredet lidt, men der er *virkelig* stadig gang i den...)



Questions

